



One Gateway, Six Clusters: Routing, Federation, and Zero-Downtime Upgrades at eBay



Huanjie Guo

Authors



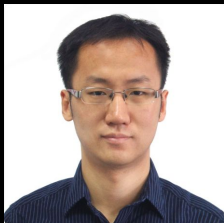
Takawa Tian
Engineer



Huanjie Guo
Engineer



Jinyang Li
Engineer



Gang Yin
Airflow Tech Lead



Bin Wei
Manager



Jiaye Wang
Manager

Agenda



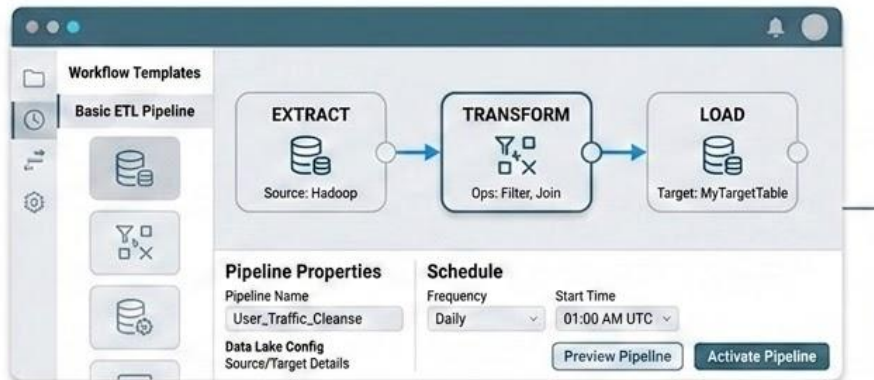
- **Background**
- **Gateway Design**
- **Zero Downtime - Airflow Upgrade**
- **Concurrency Control**
- **What's Next**
- **Q & A**

Background

eBay Data Lake: Simplified ETL & Scheduling.

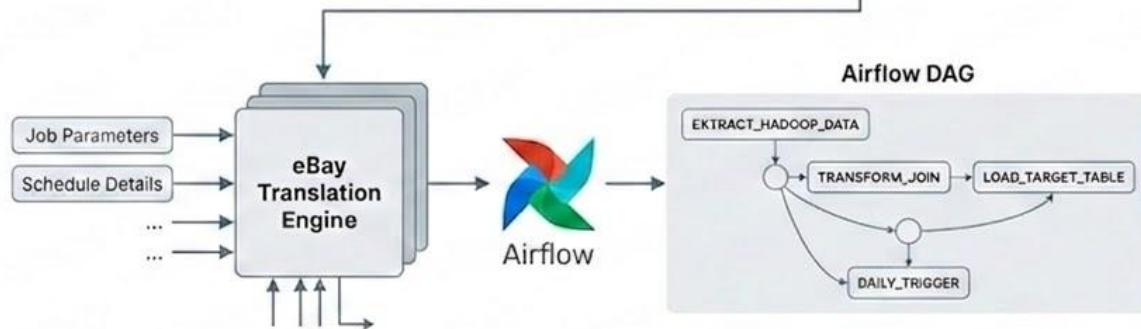


Data Lake Platform UI (User View)

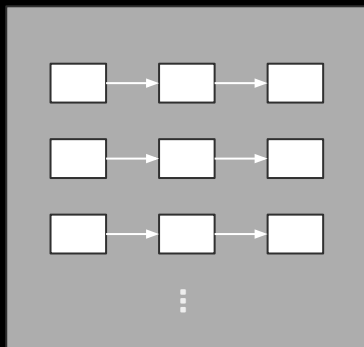


Automated Airflow Translation & Scheduling (Backend Flow)

User configurations are converted into native Airflow Python scripts (DAGs).



Background



1000 Pipelines

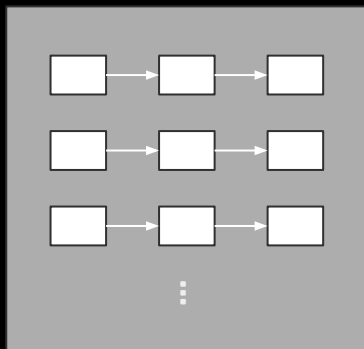


Single Airflow Cluster



System Performance: **STABLE**

Background



10000+ Pipelines



Single Airflow Cluster



System Performance: **UNSTABLE**

Example 1. - DB or disk issue



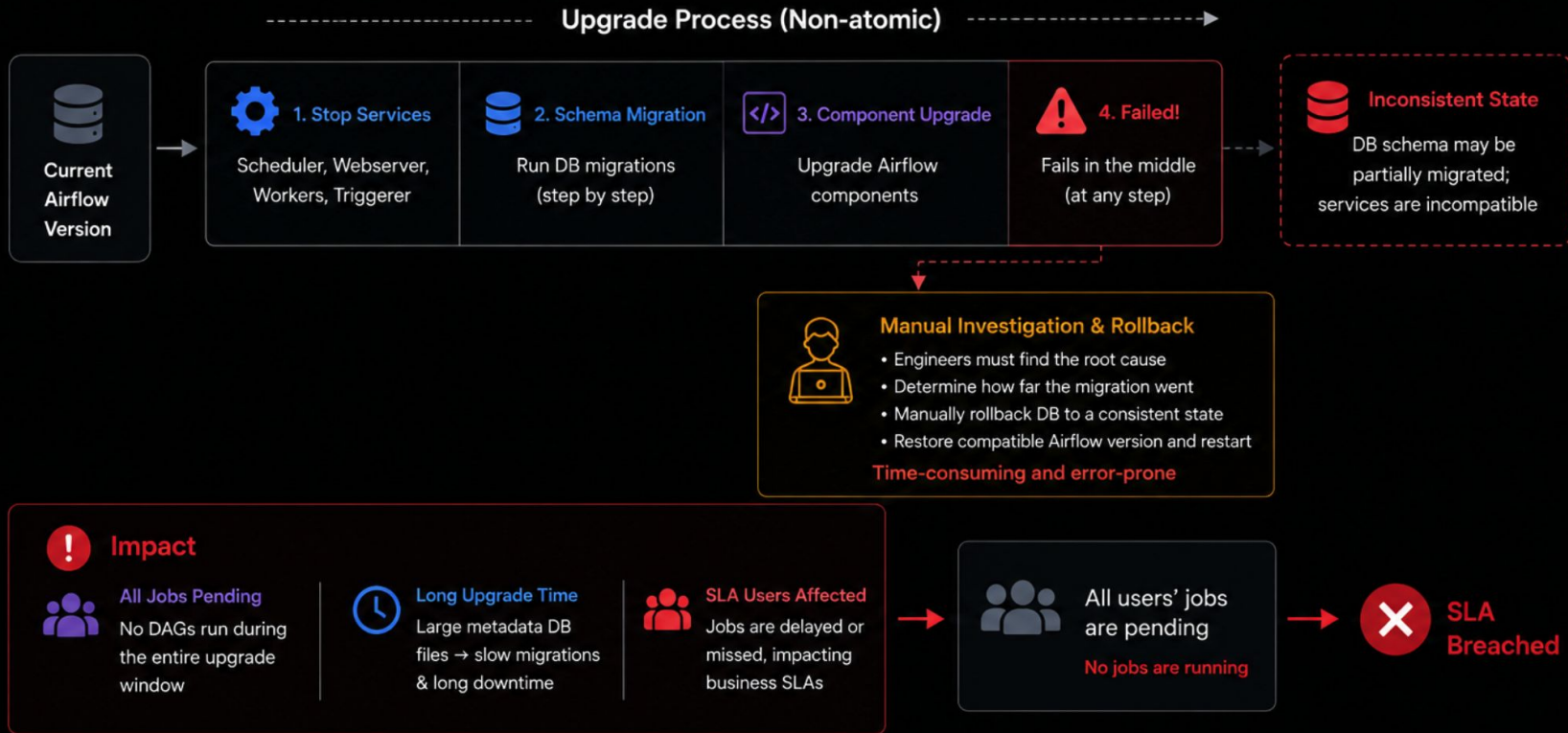
IMPACT

- Task Execution Delays
- Data Sync Inconsistency
- Slow Service Response

Example 2. Non-atomic Airflow Upgrade

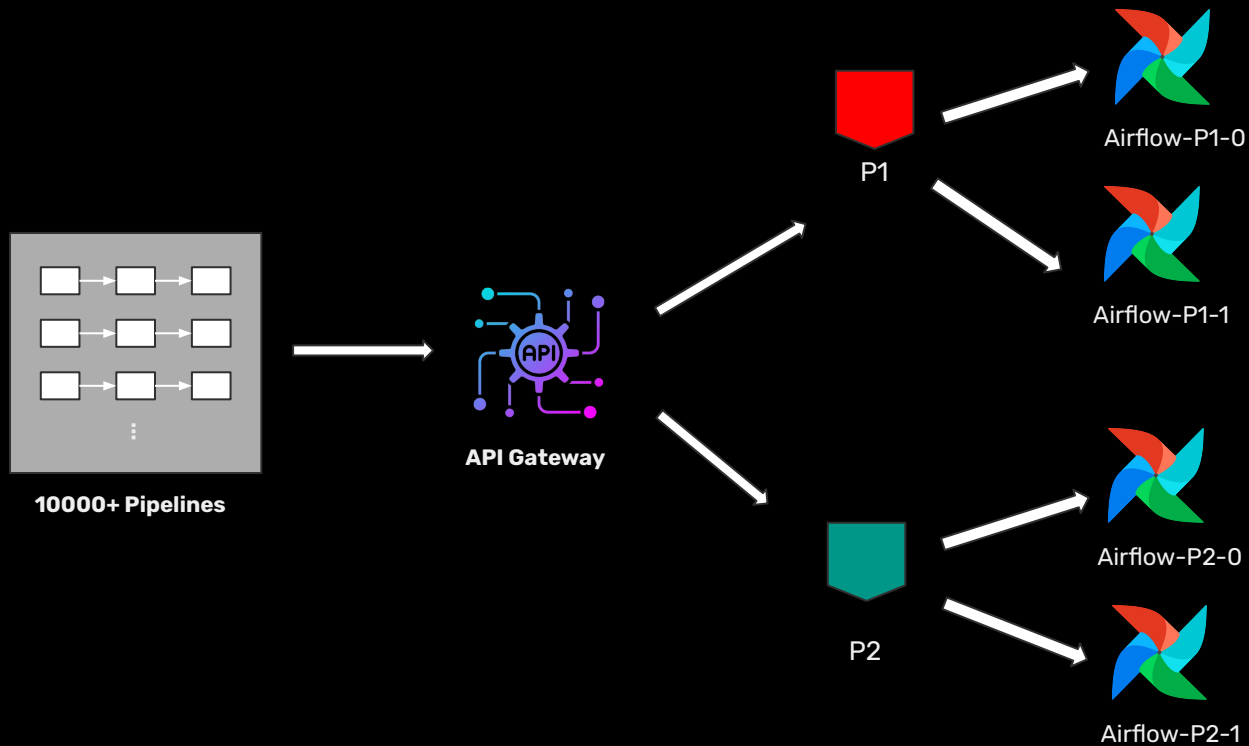


Upgrade is **not atomic**. It may fail midway and require manual intervention.

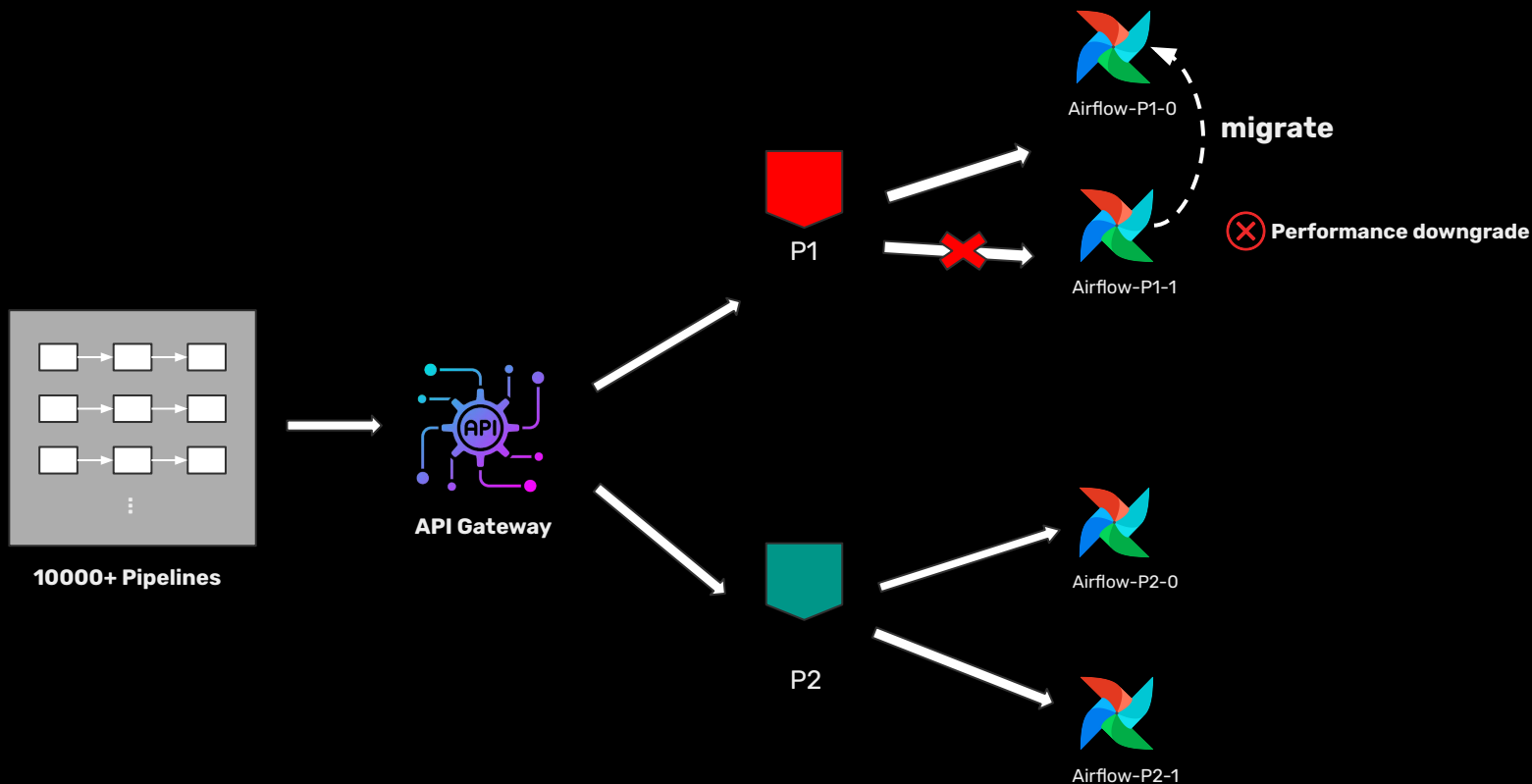


Airflow Gateway

Gateway

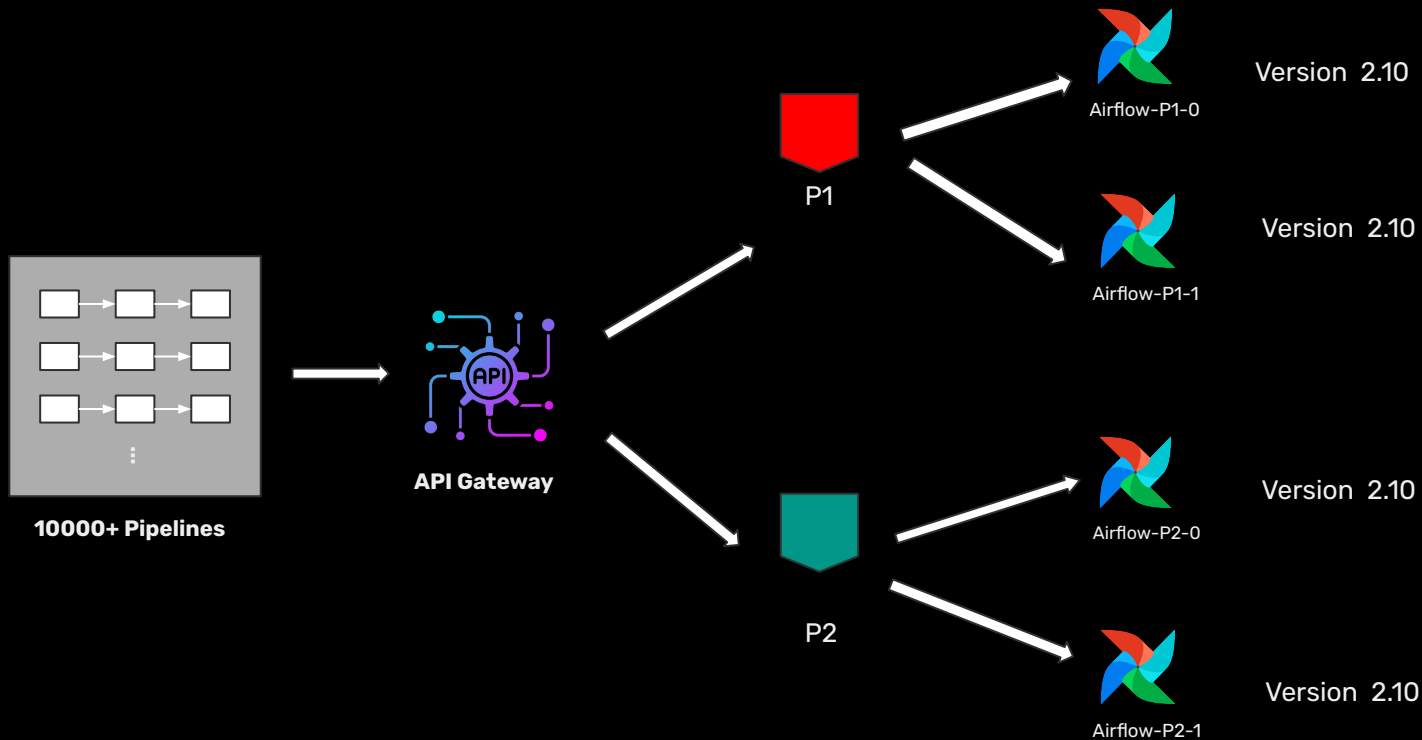


Gateway - Cluster Issue Handling

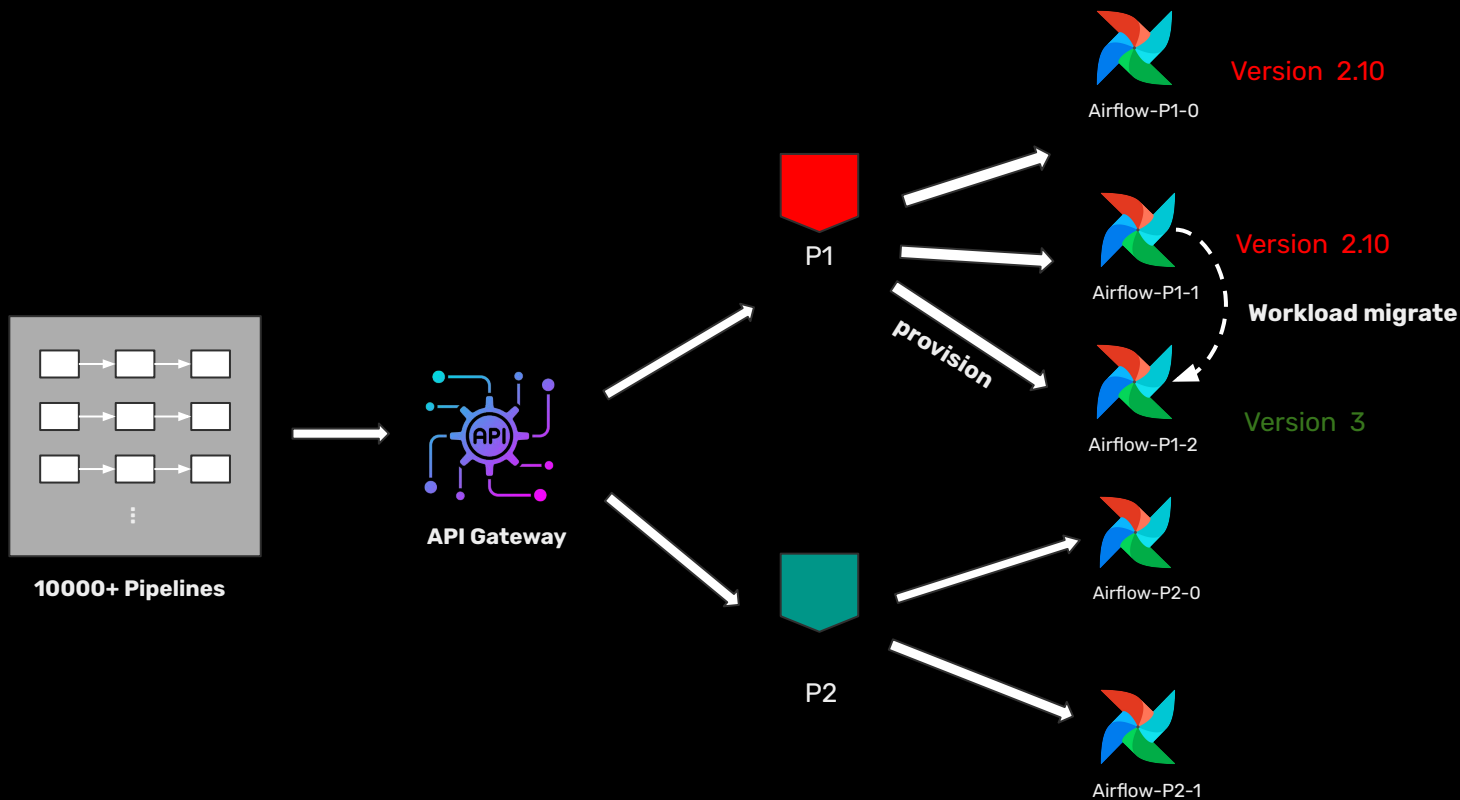


Zero-Downtime Upgrade

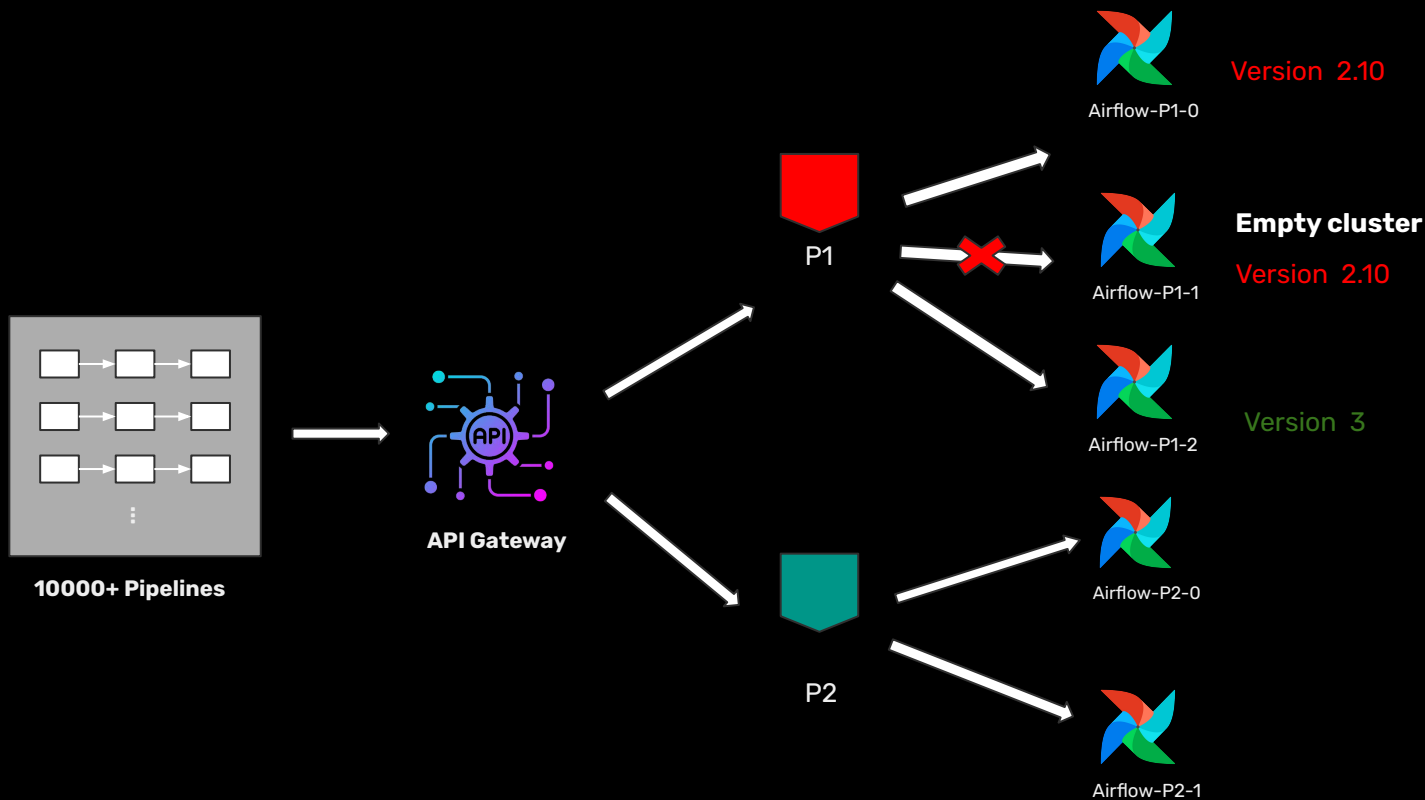
Zero-downtime upgrade



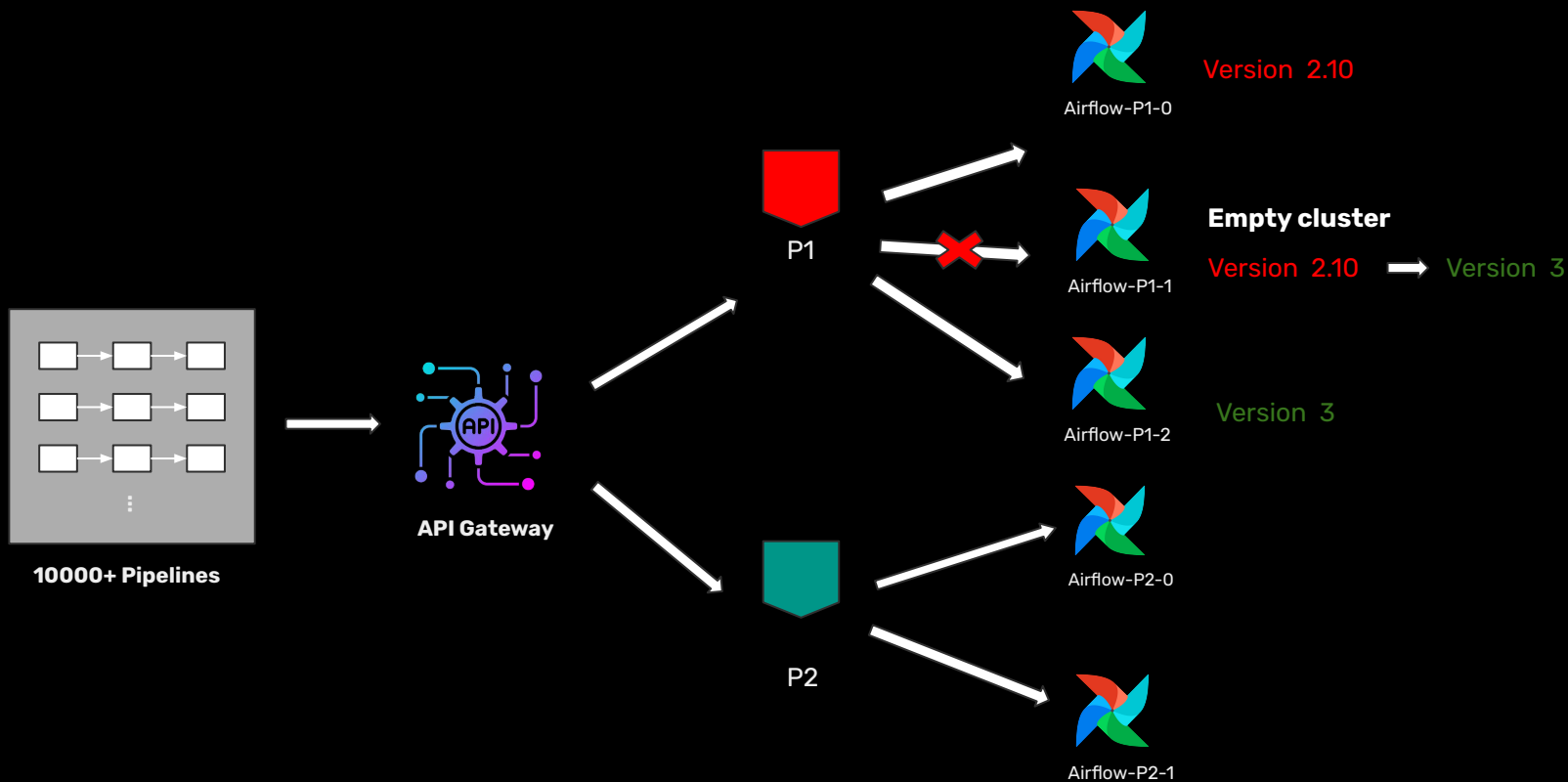
Zero-downtime upgrade



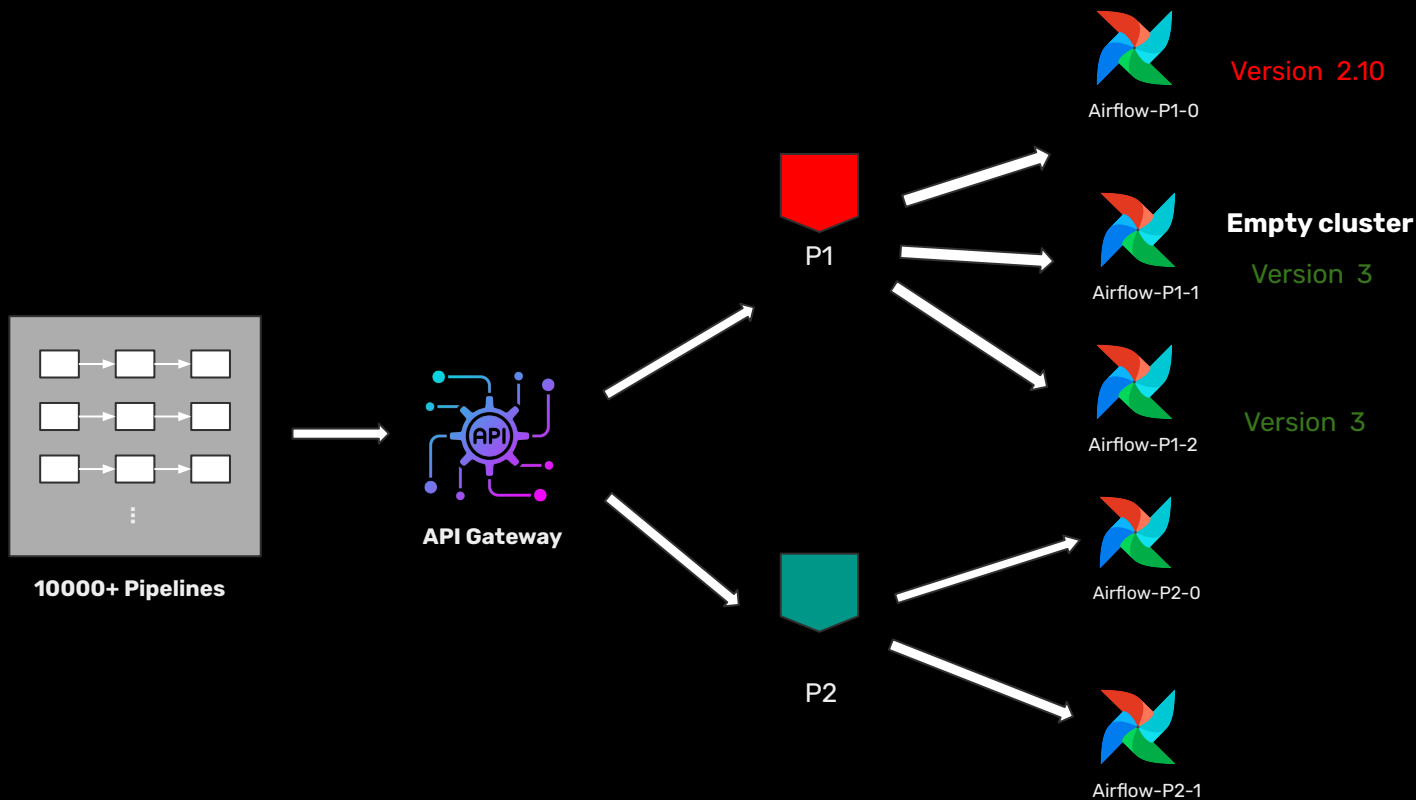
Zero-downtime upgrade



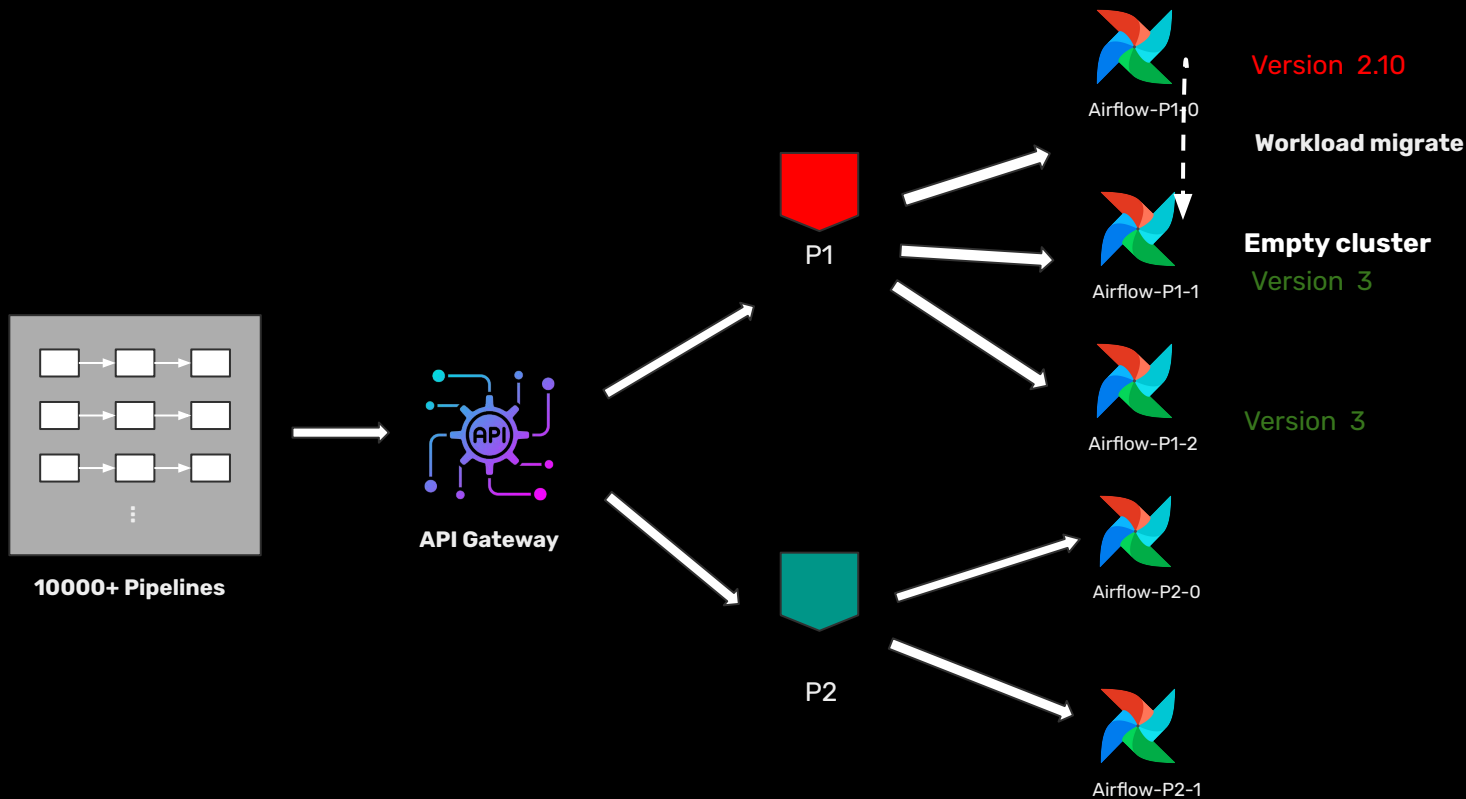
Zero-downtime upgrade



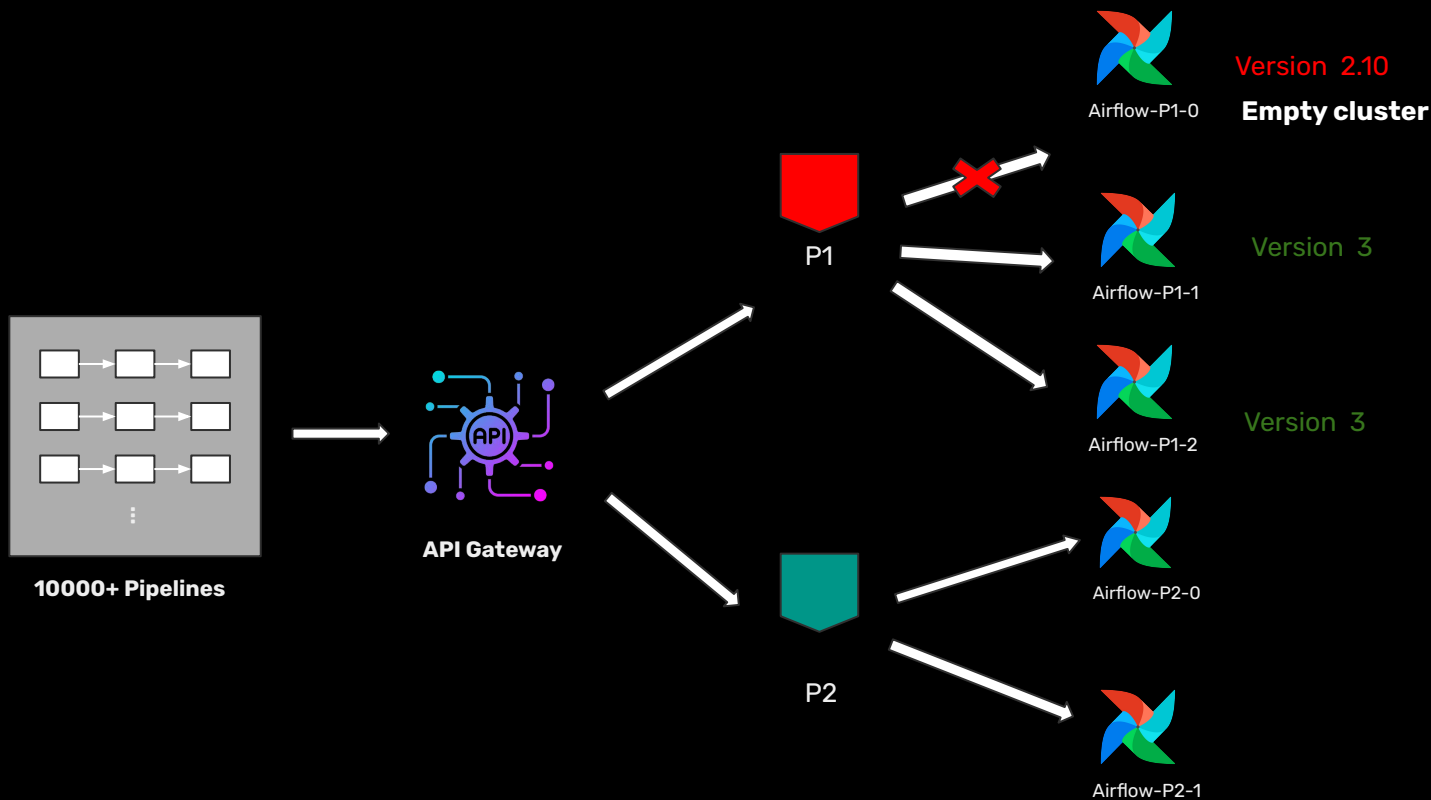
Zero-downtime upgrade



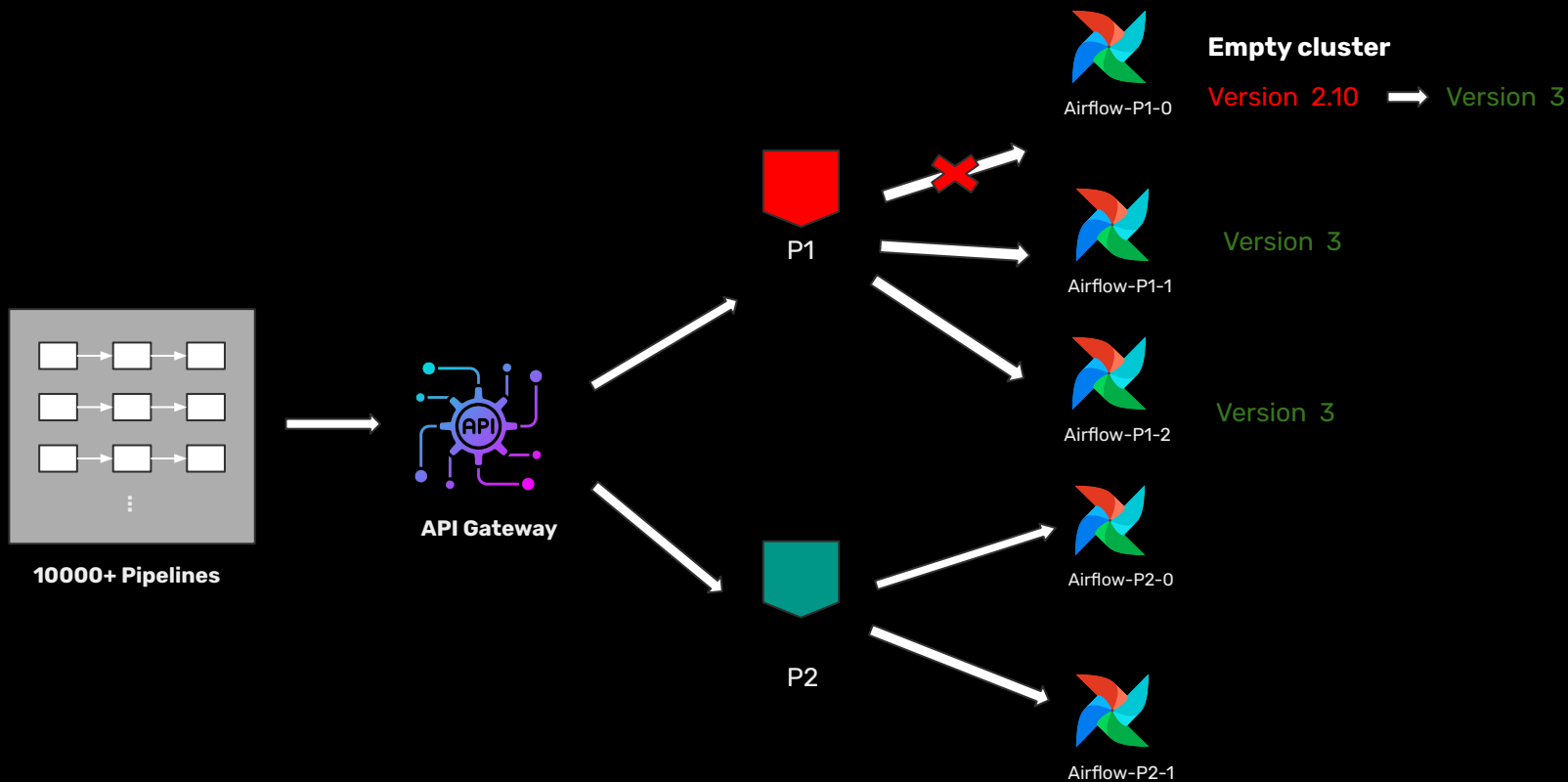
Zero-downtime upgrade



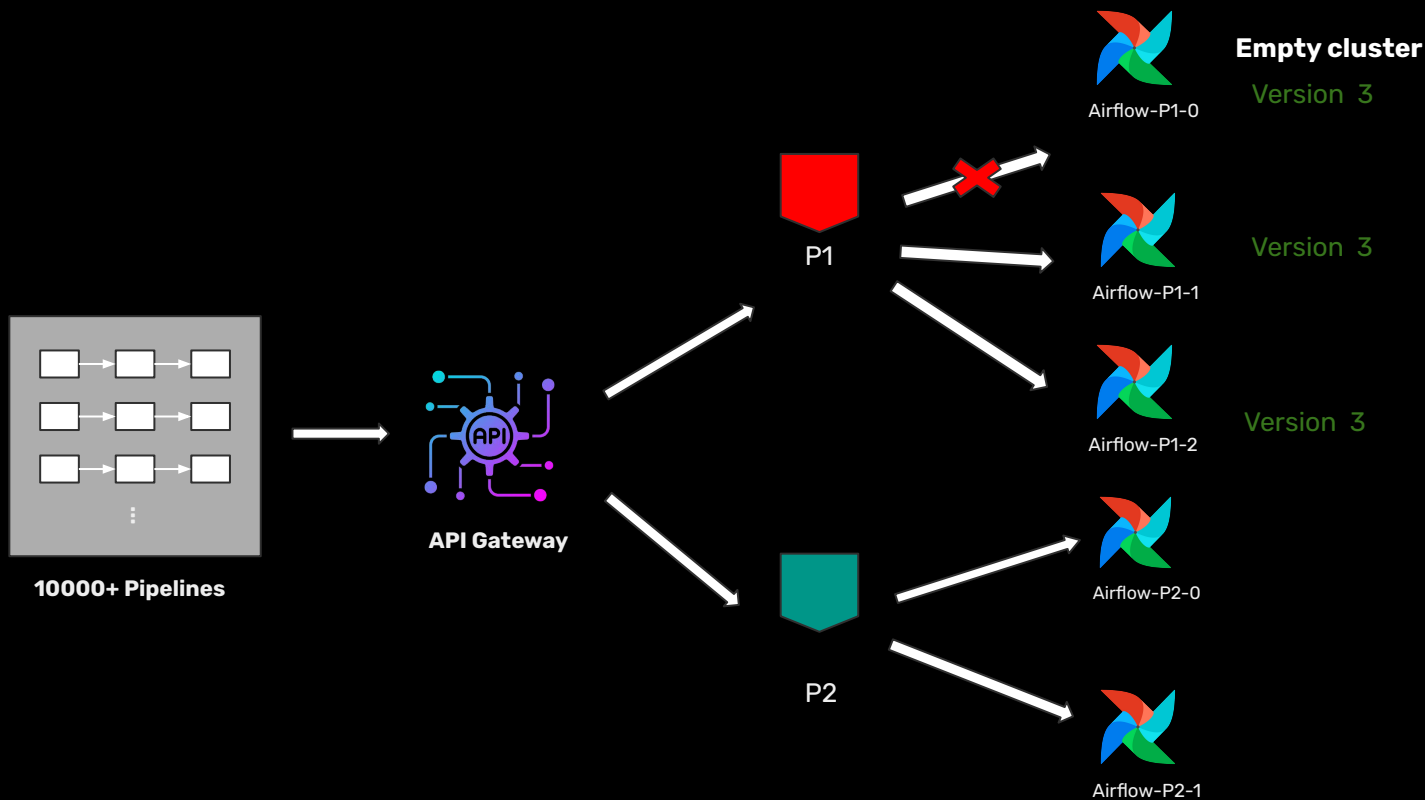
Zero-downtime upgrade



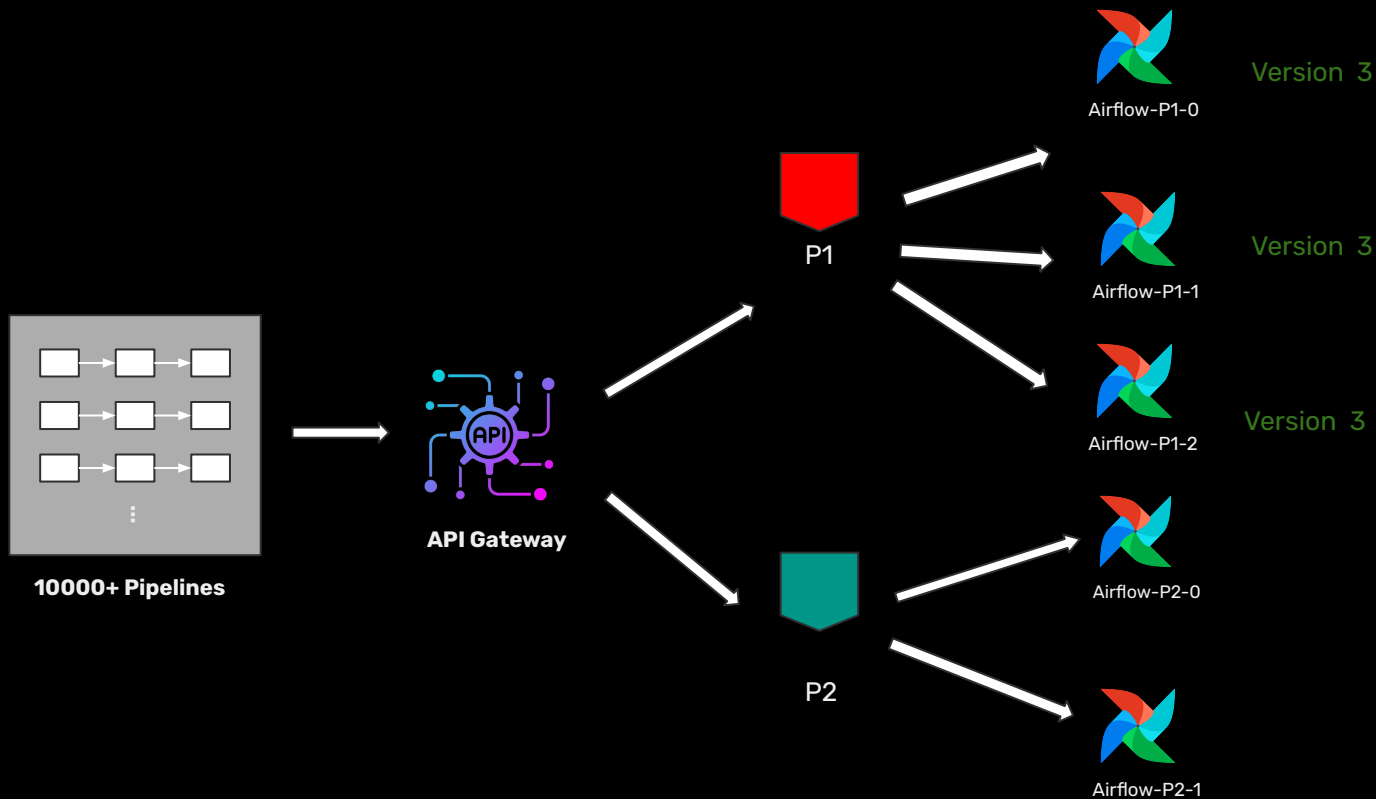
Zero-downtime upgrade



Zero-downtime upgrade

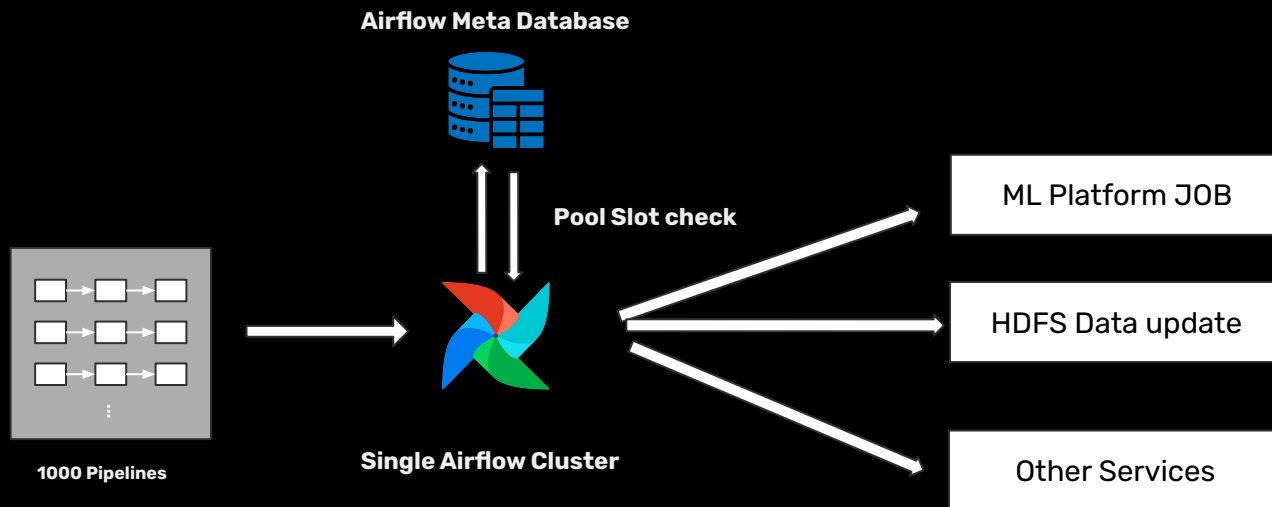


Zero-downtime upgrade

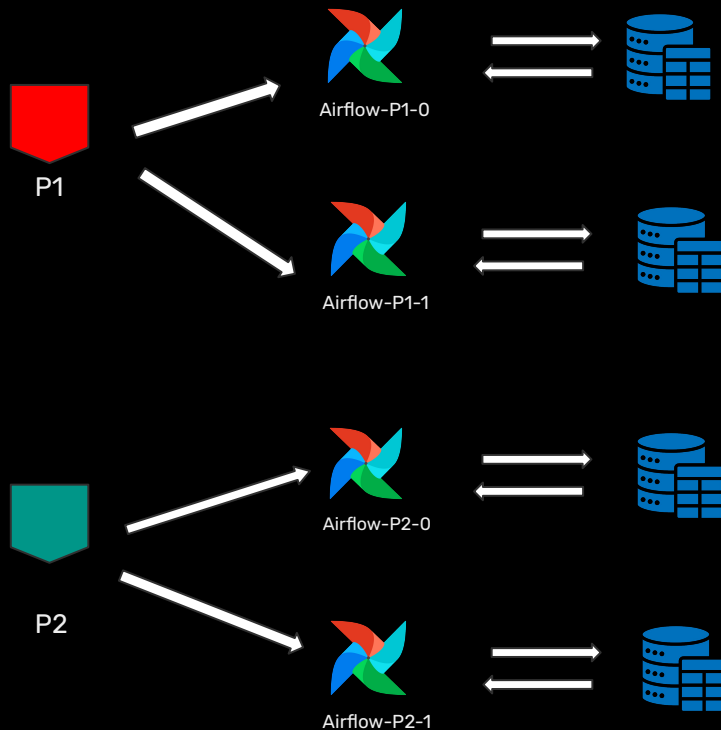


Concurrency Control

Airflow Community concurrency control: Pool

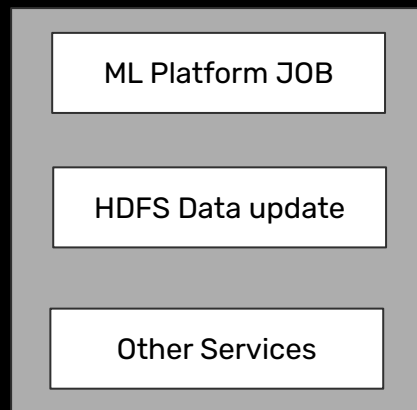


Multi-clusters concurrency control: Challenge

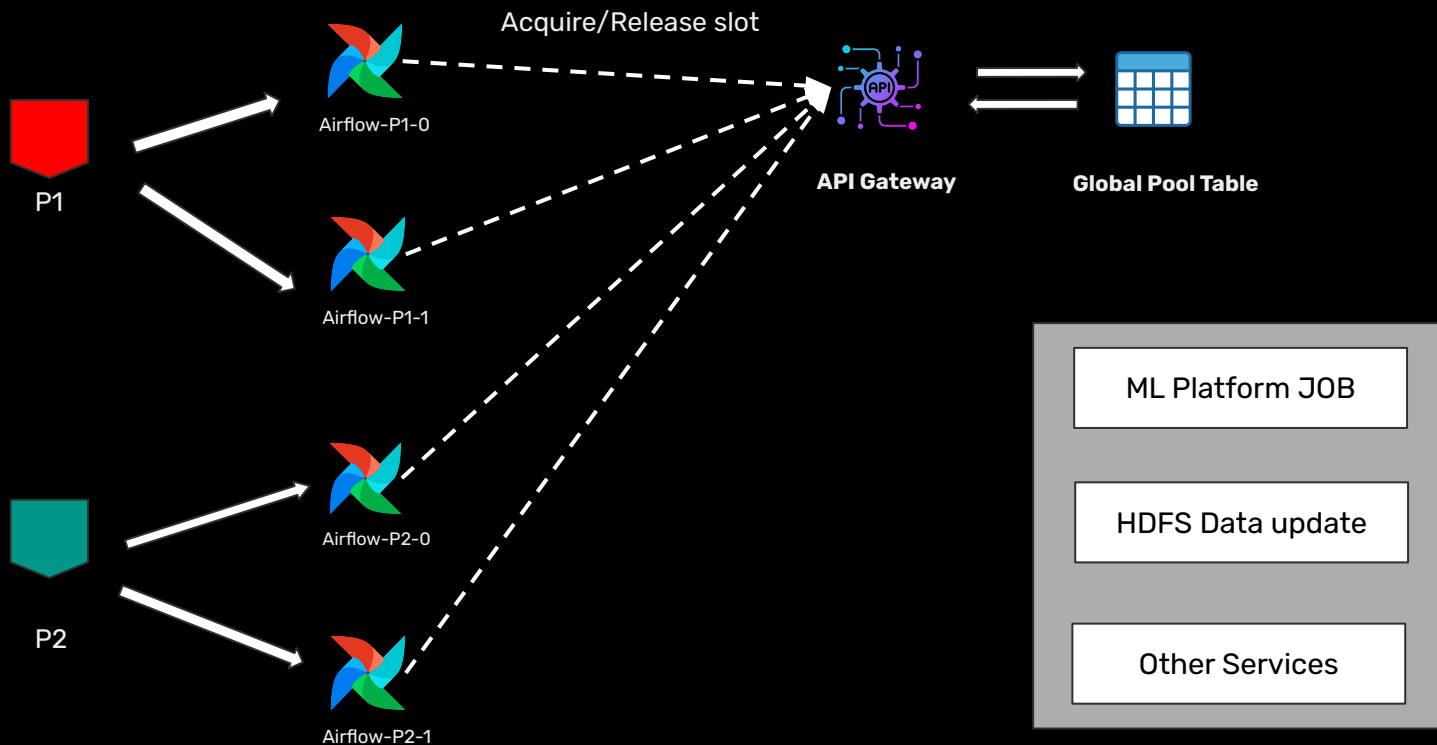


Every cluster works independently

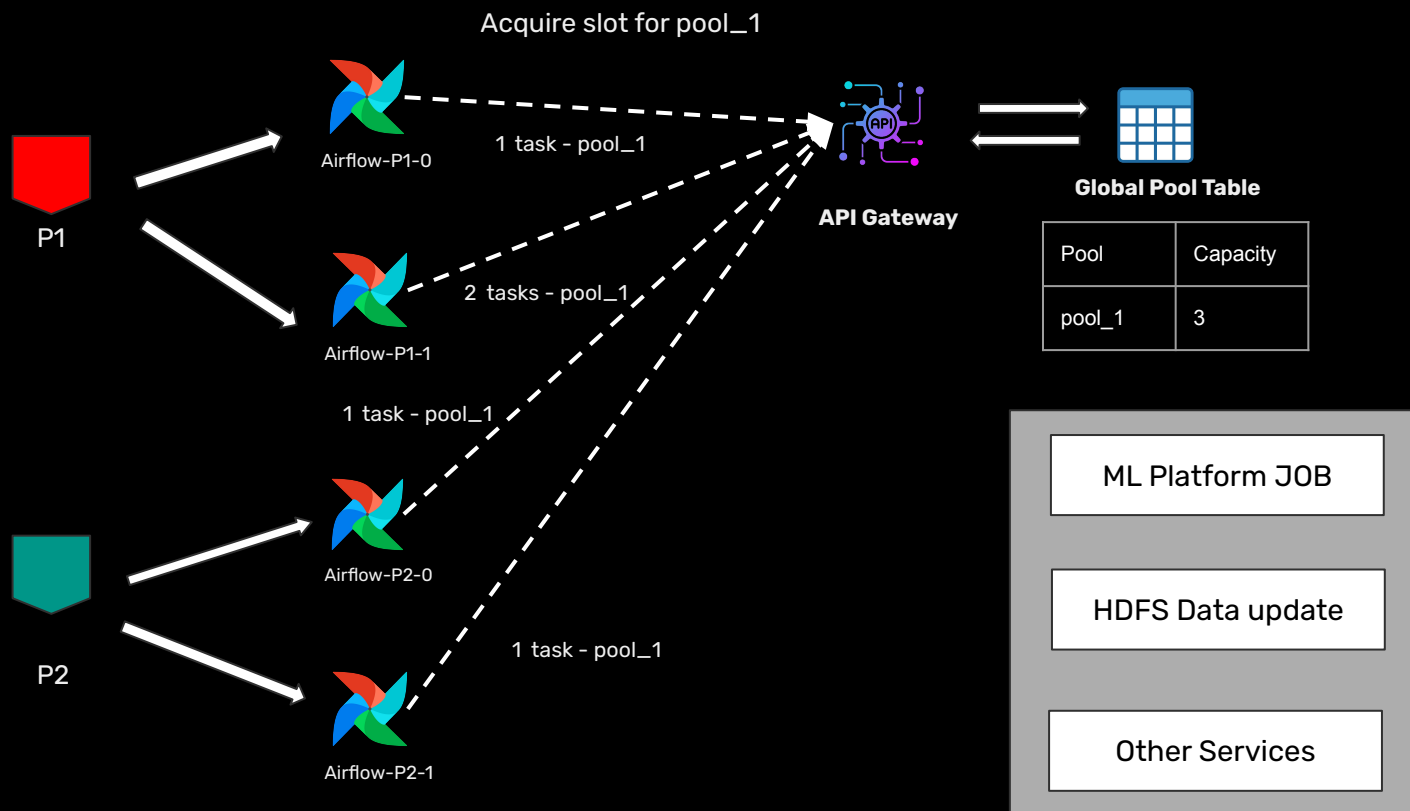
Cannot share the pool information in database



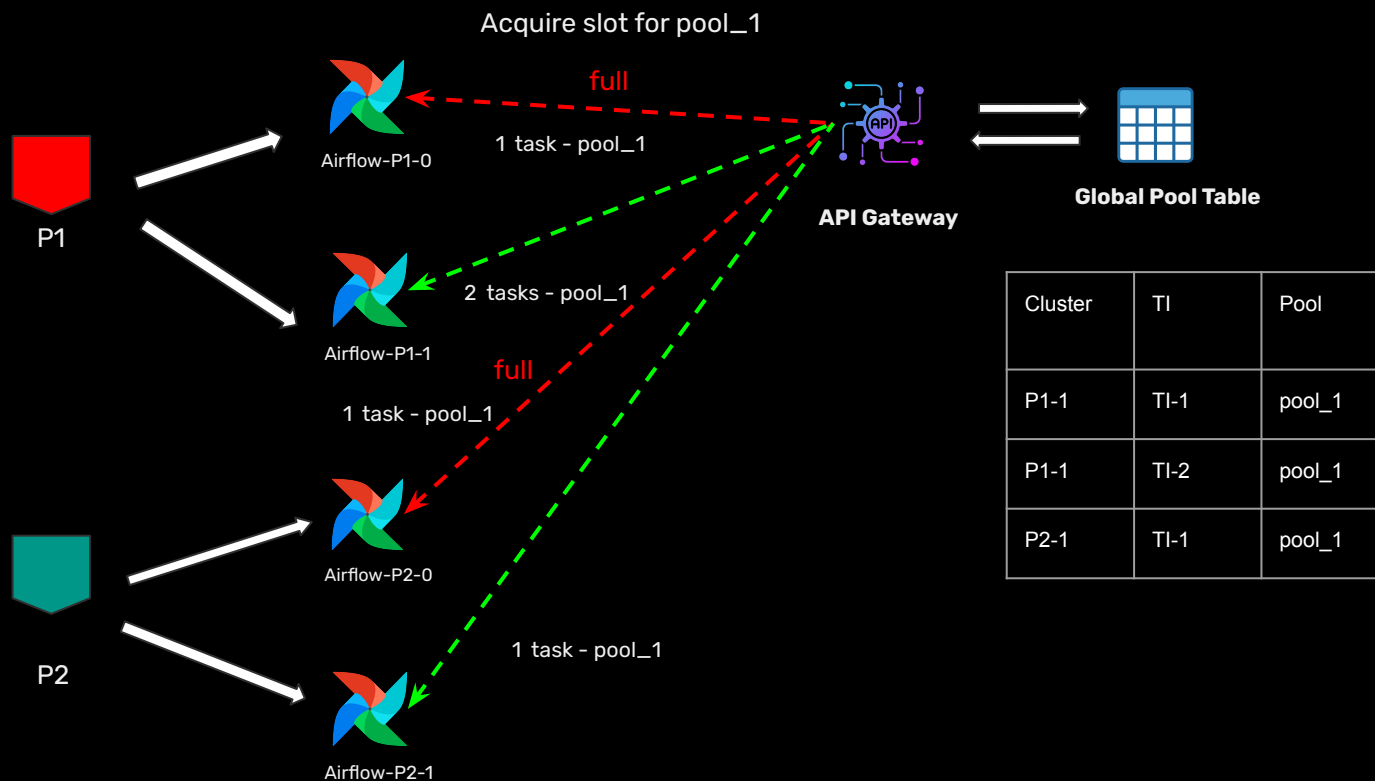
Multi-clusters concurrency control: Global Pool



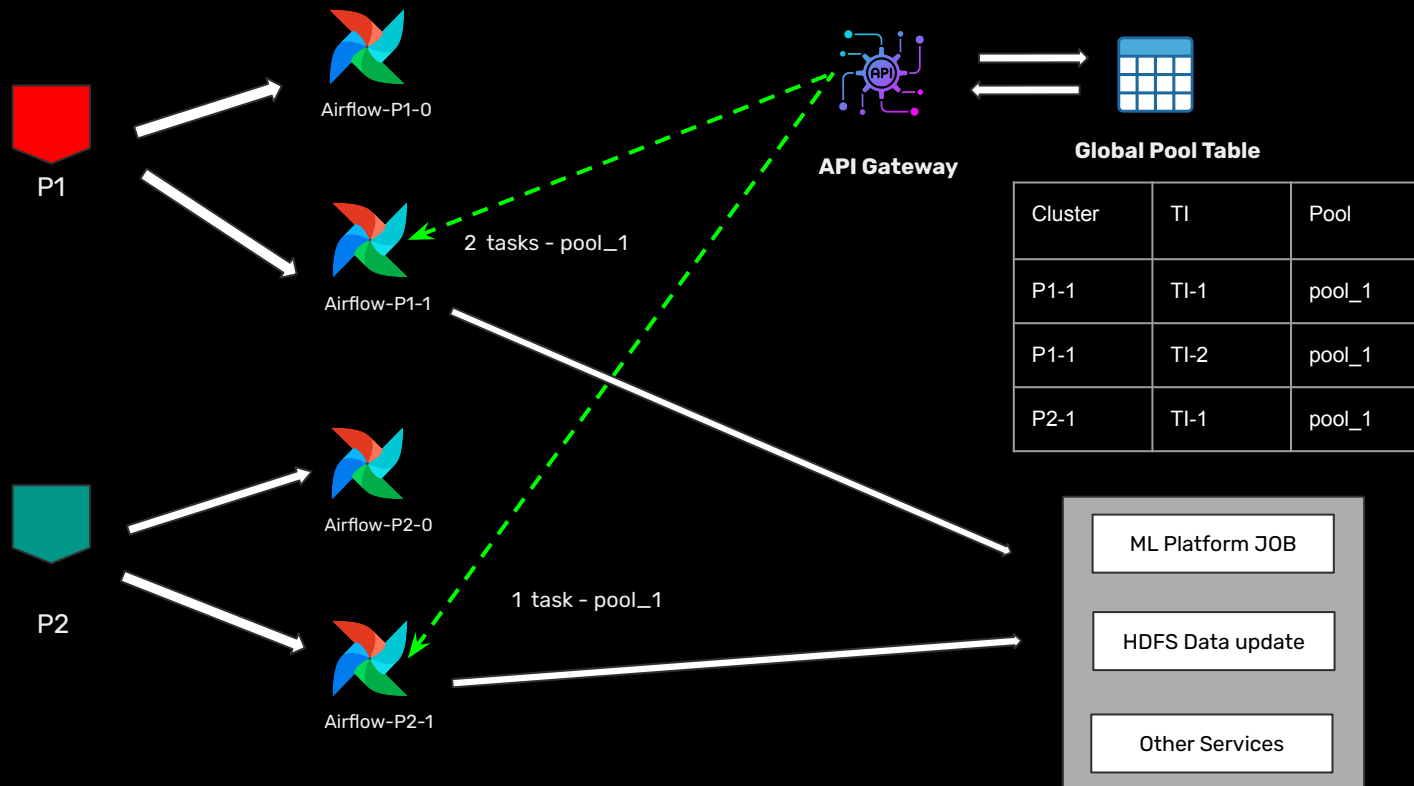
Global Pool - Acquire slot



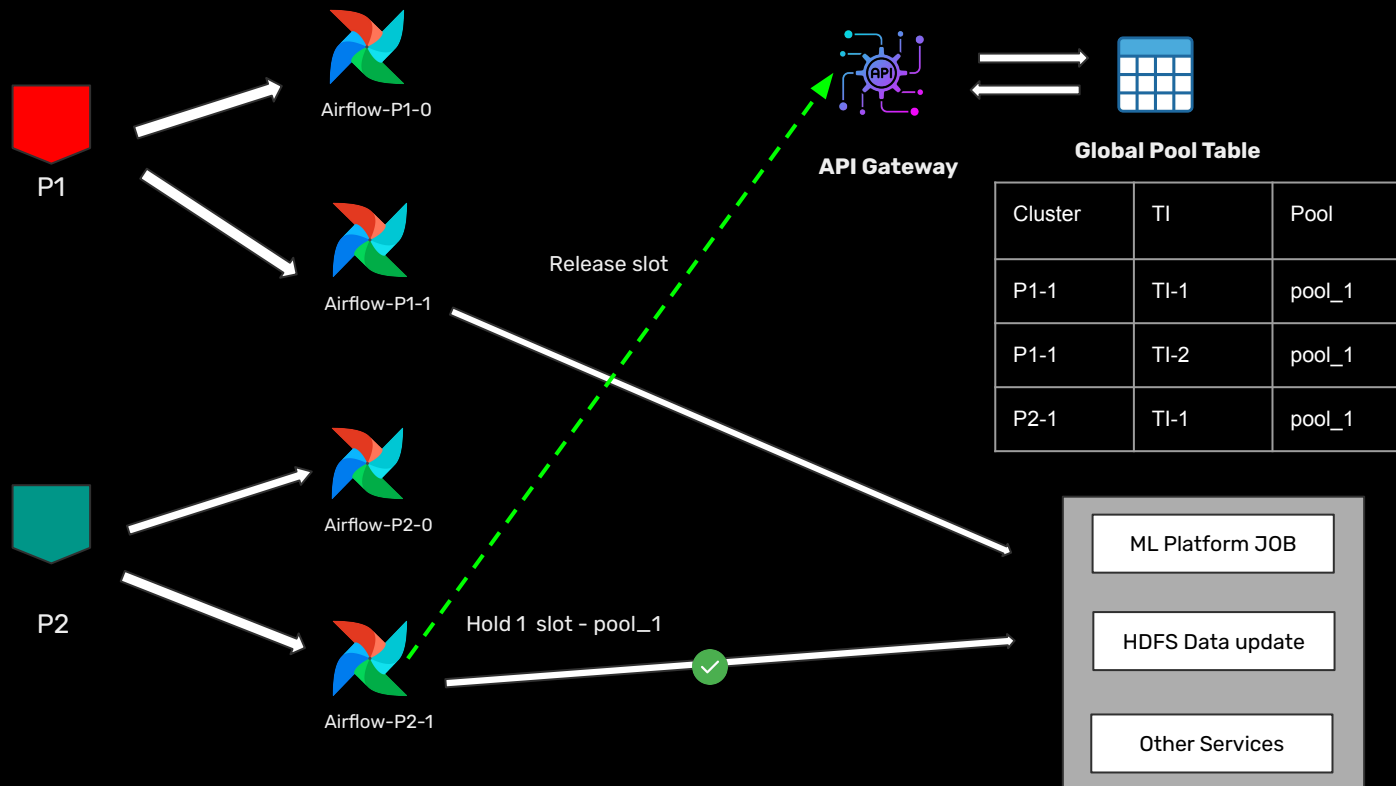
Global Pool - Acquire slot



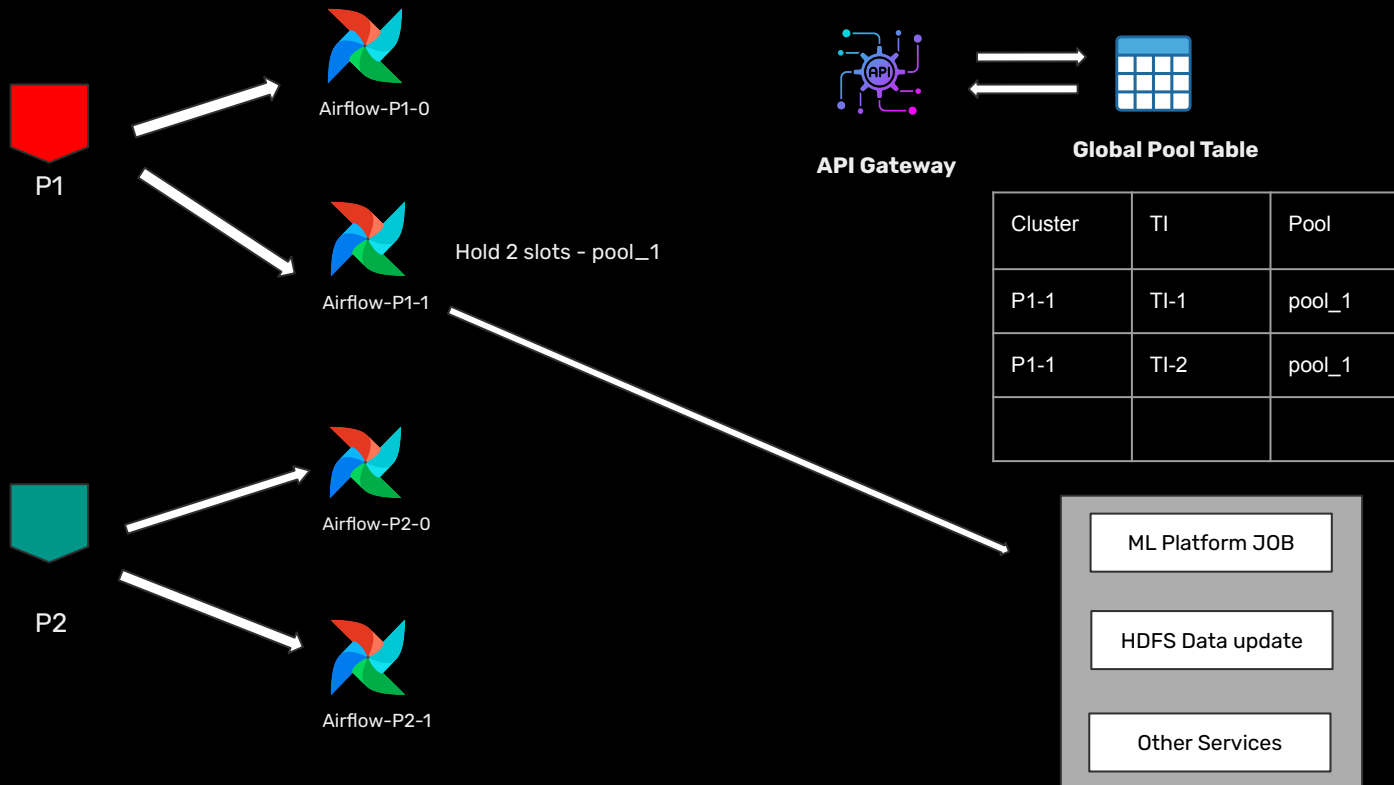
Global Pool - Hold slot and run task



Global Pool - Release slot when task finished

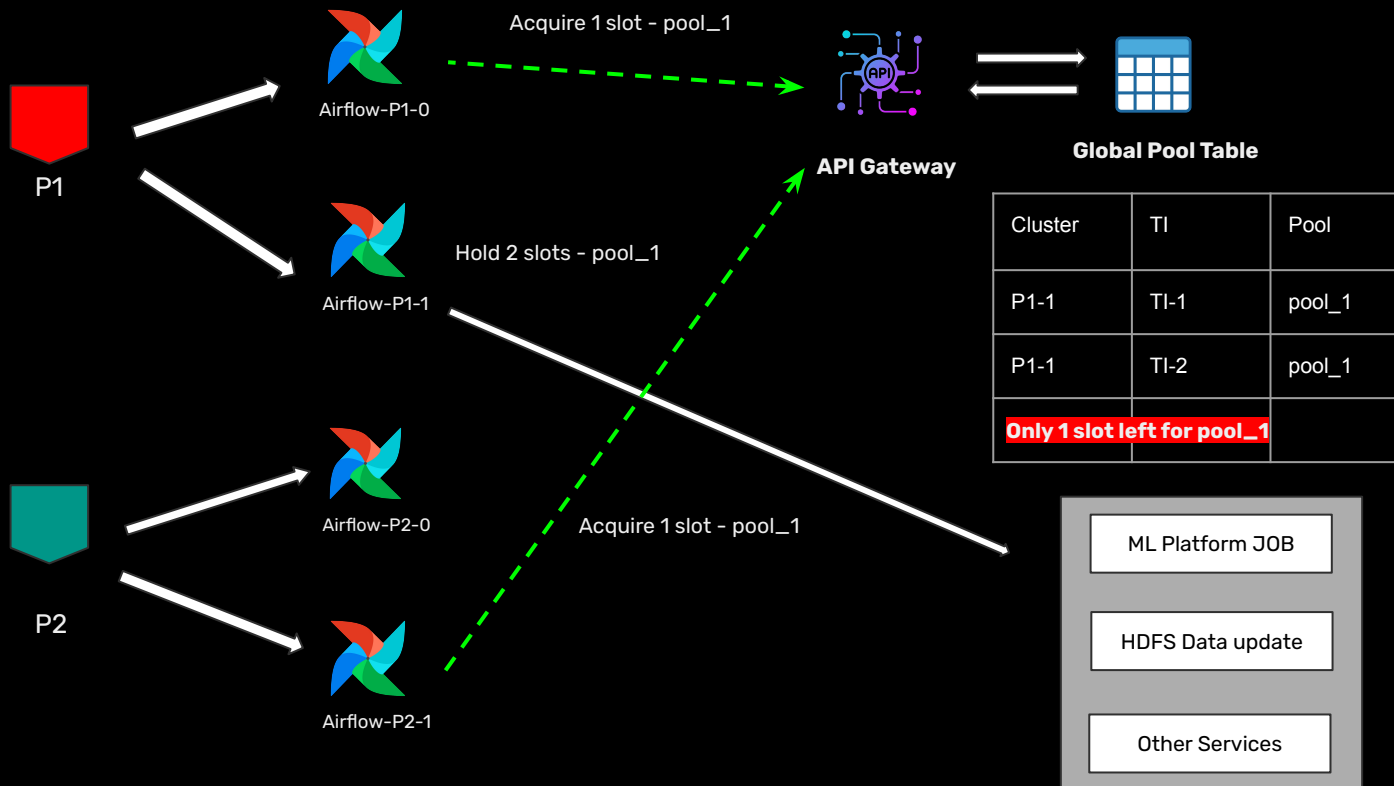


Global Pool - Release slot when task finished



How to allocate new released slots?

Global Pool - Slot scheduling

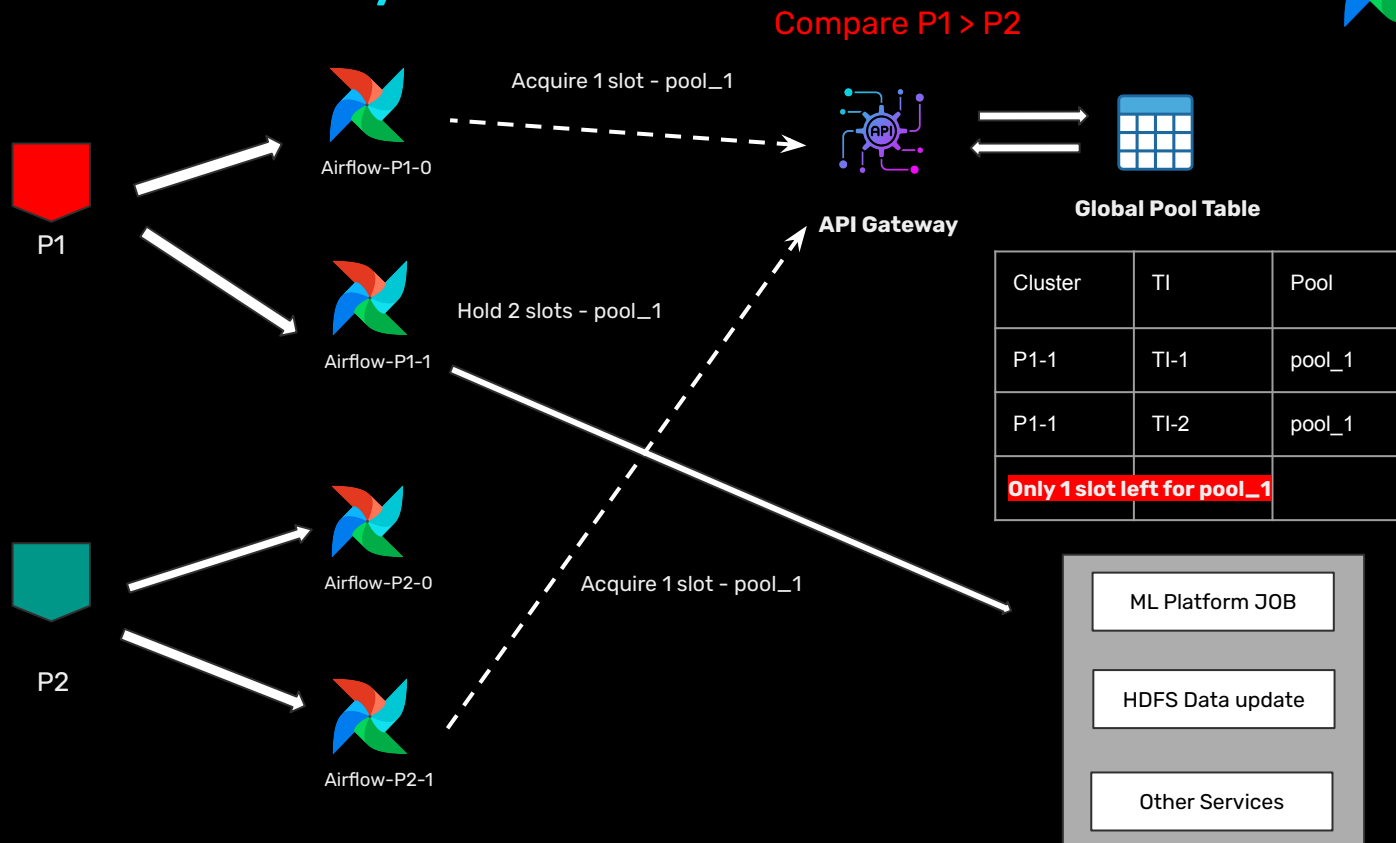


Global Pool - Slot scheduling algorithm

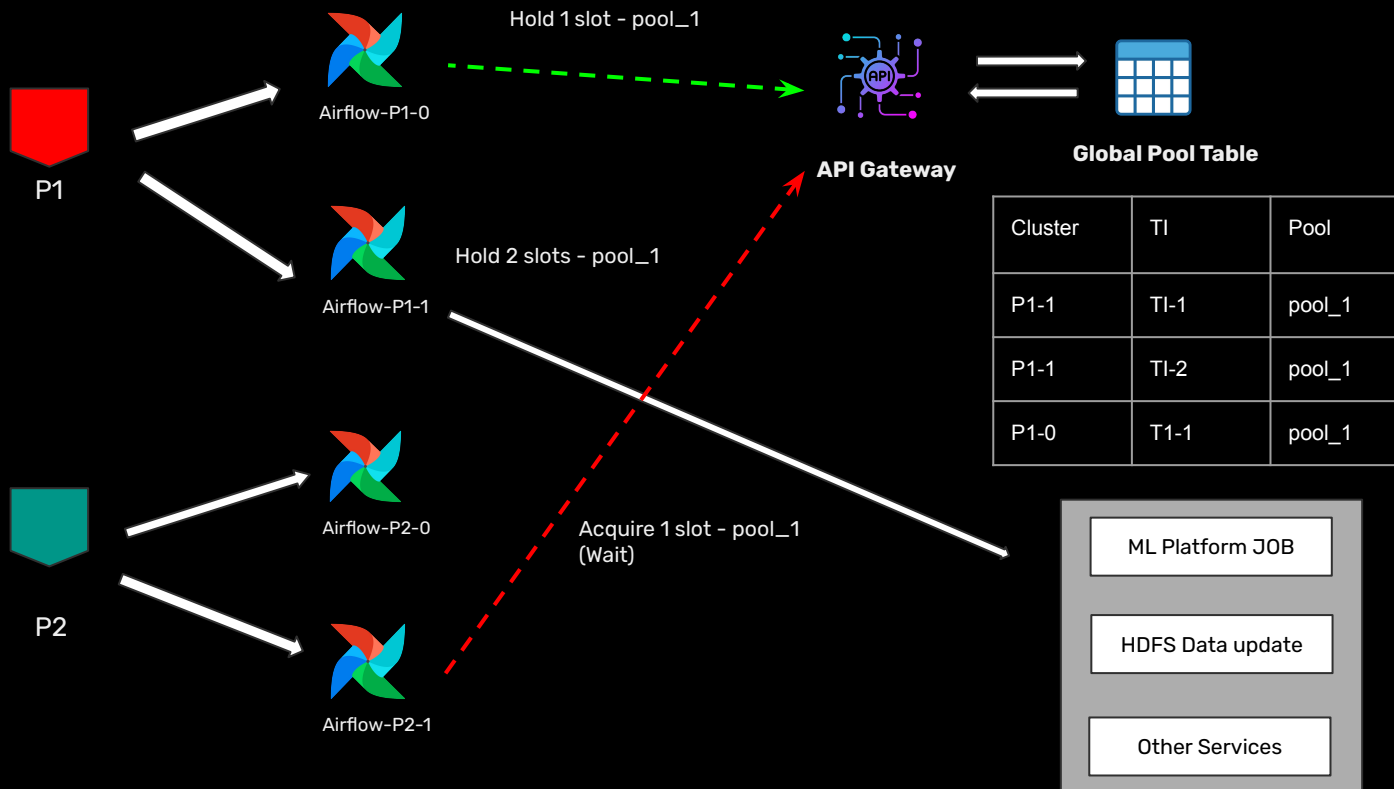


- **Random**
- **Short Job First (SJF)**
- **First In First Out (FIFO)**
- **Priority Queue (PQ) ***
- **Multiple Level Feedback Queue (MLFQ) ***

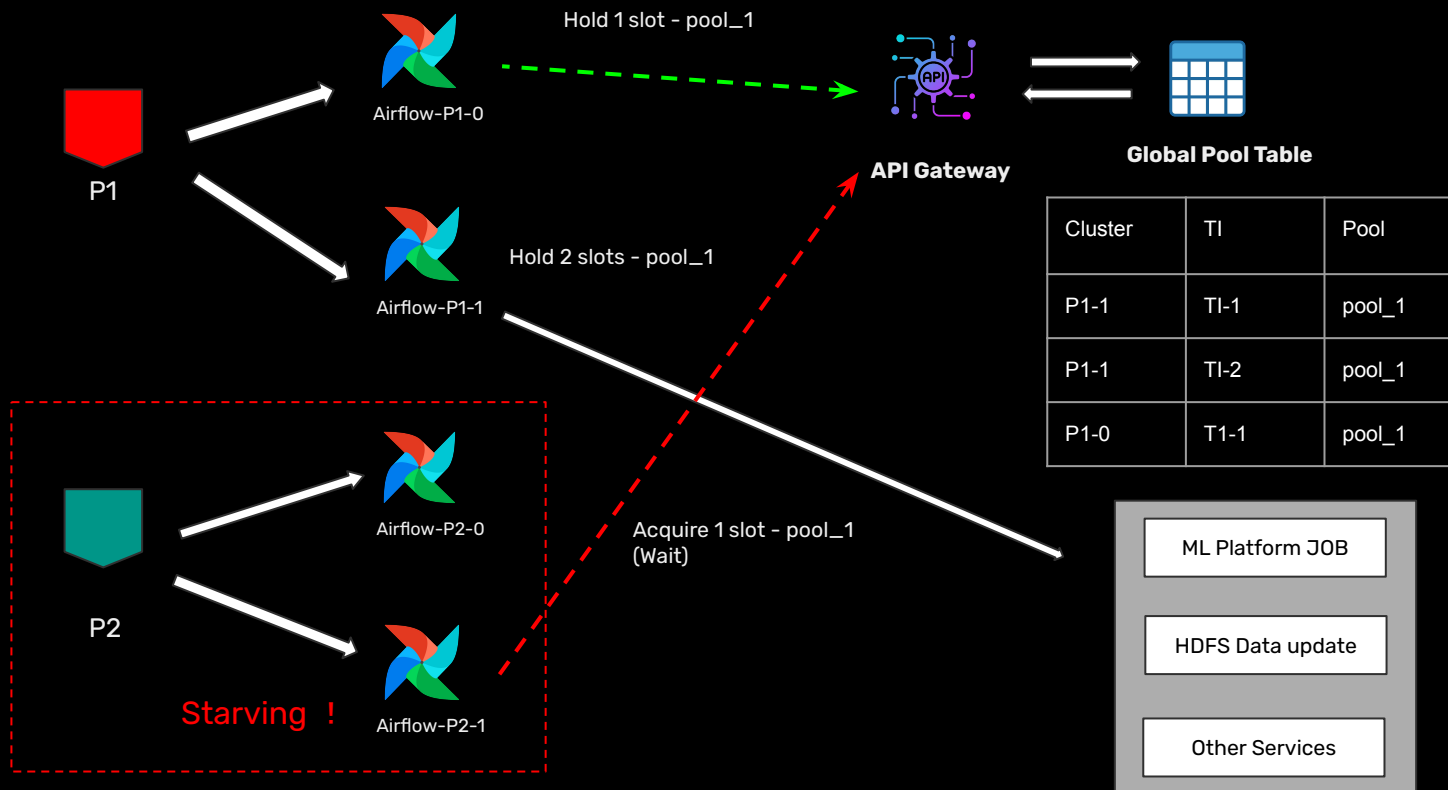
Global Pool - Priority Queue



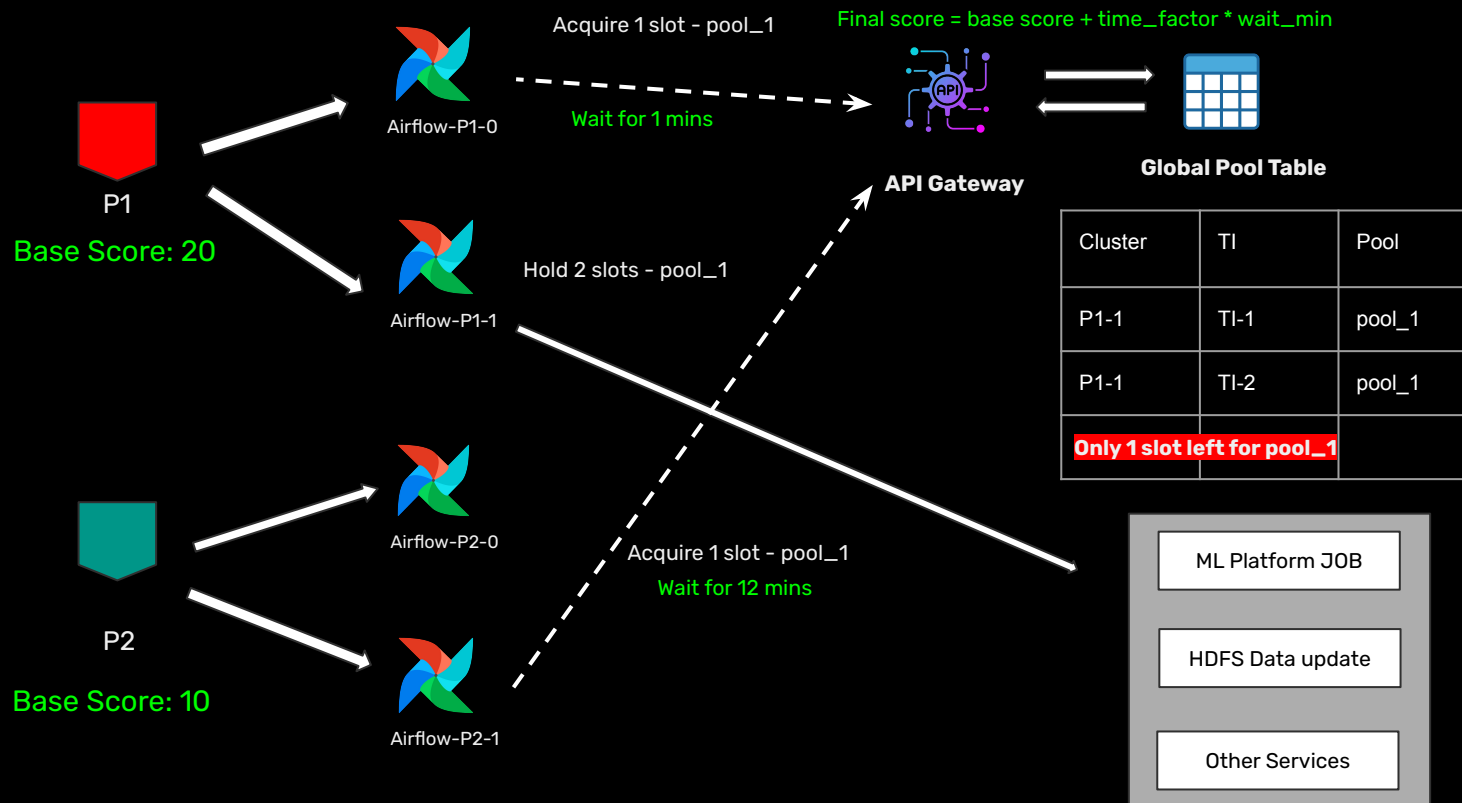
Global Pool - Priority Queue



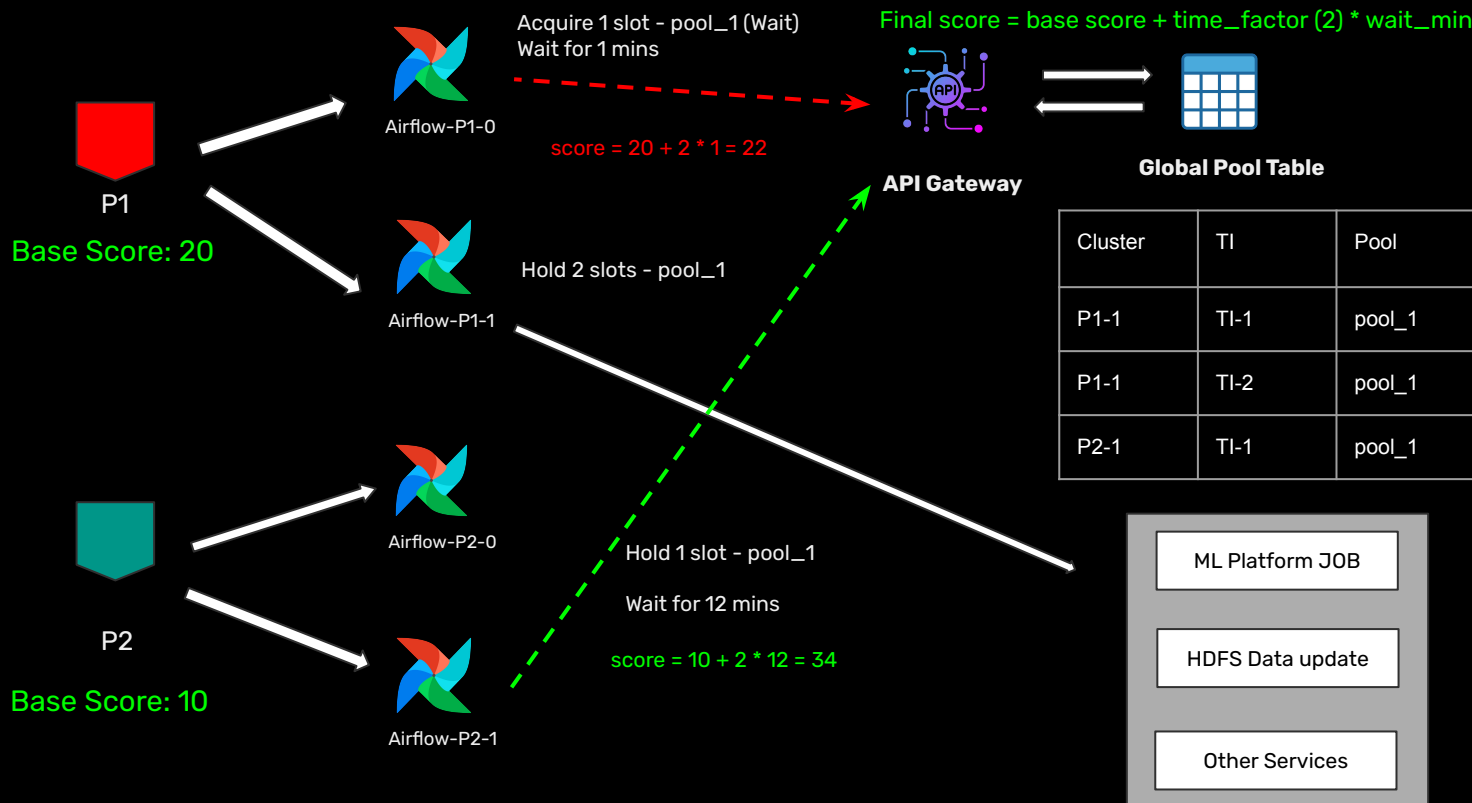
Global Pool - Priority Queue



Global Pool - Multiple Level Feedback Queue

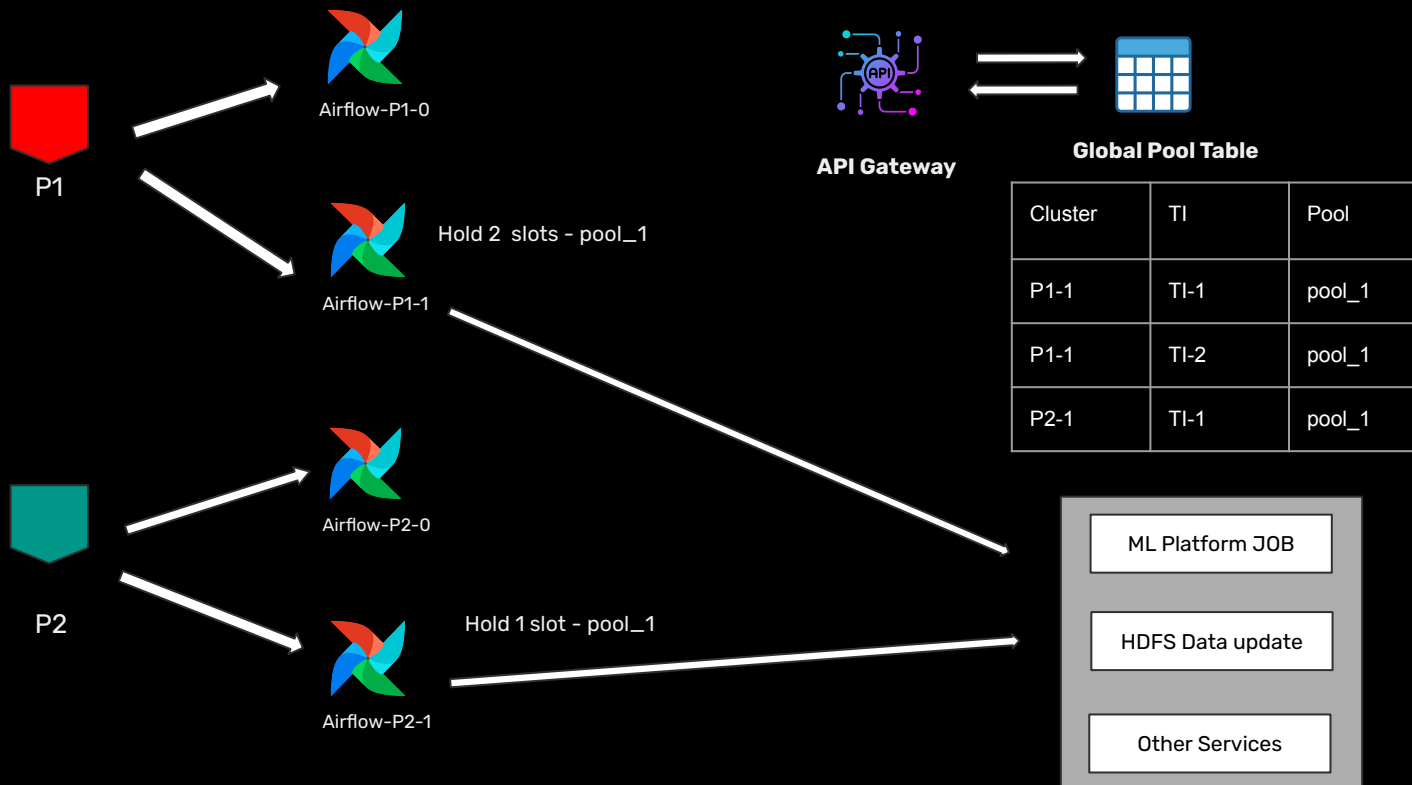


Global Pool - Multiple Level Feedback Queue

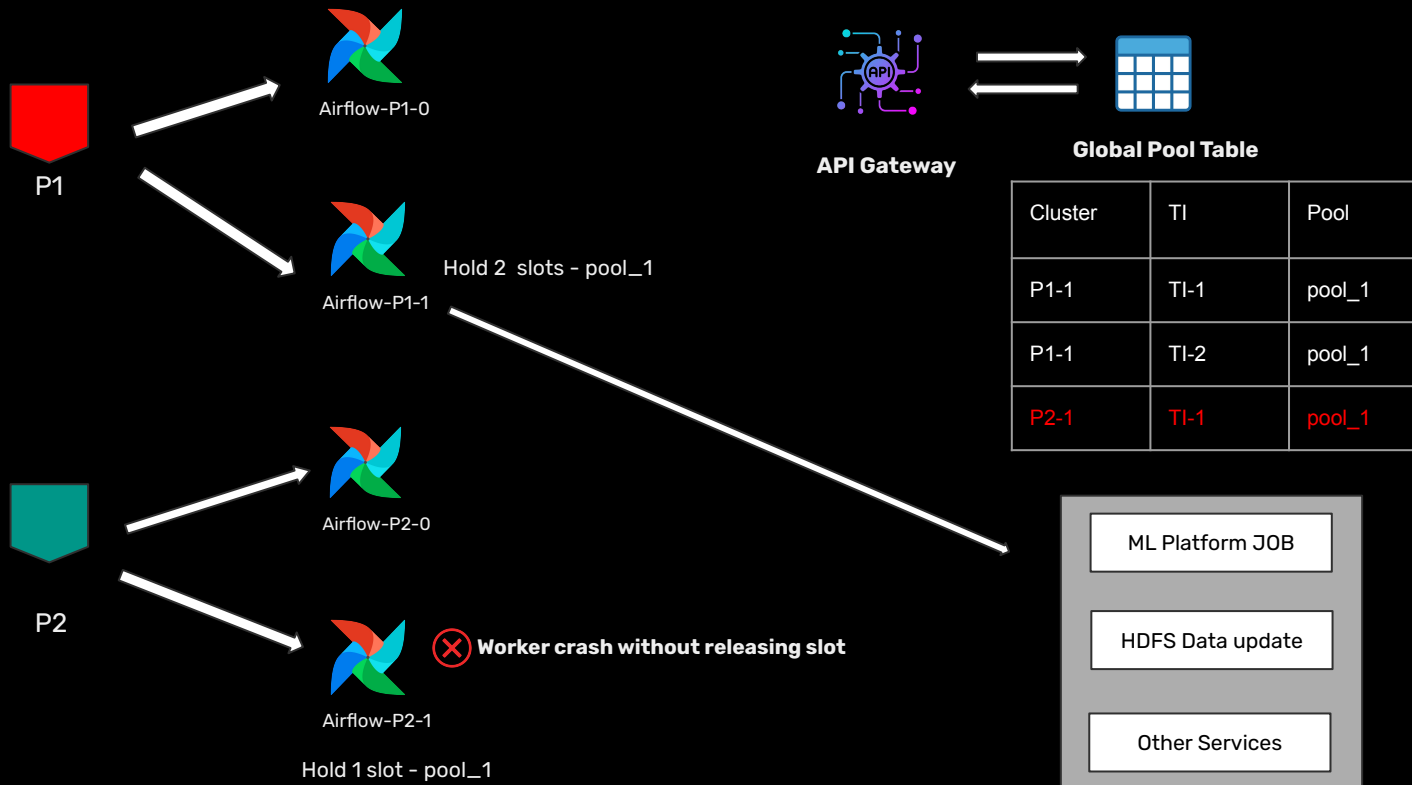


Exception Handling

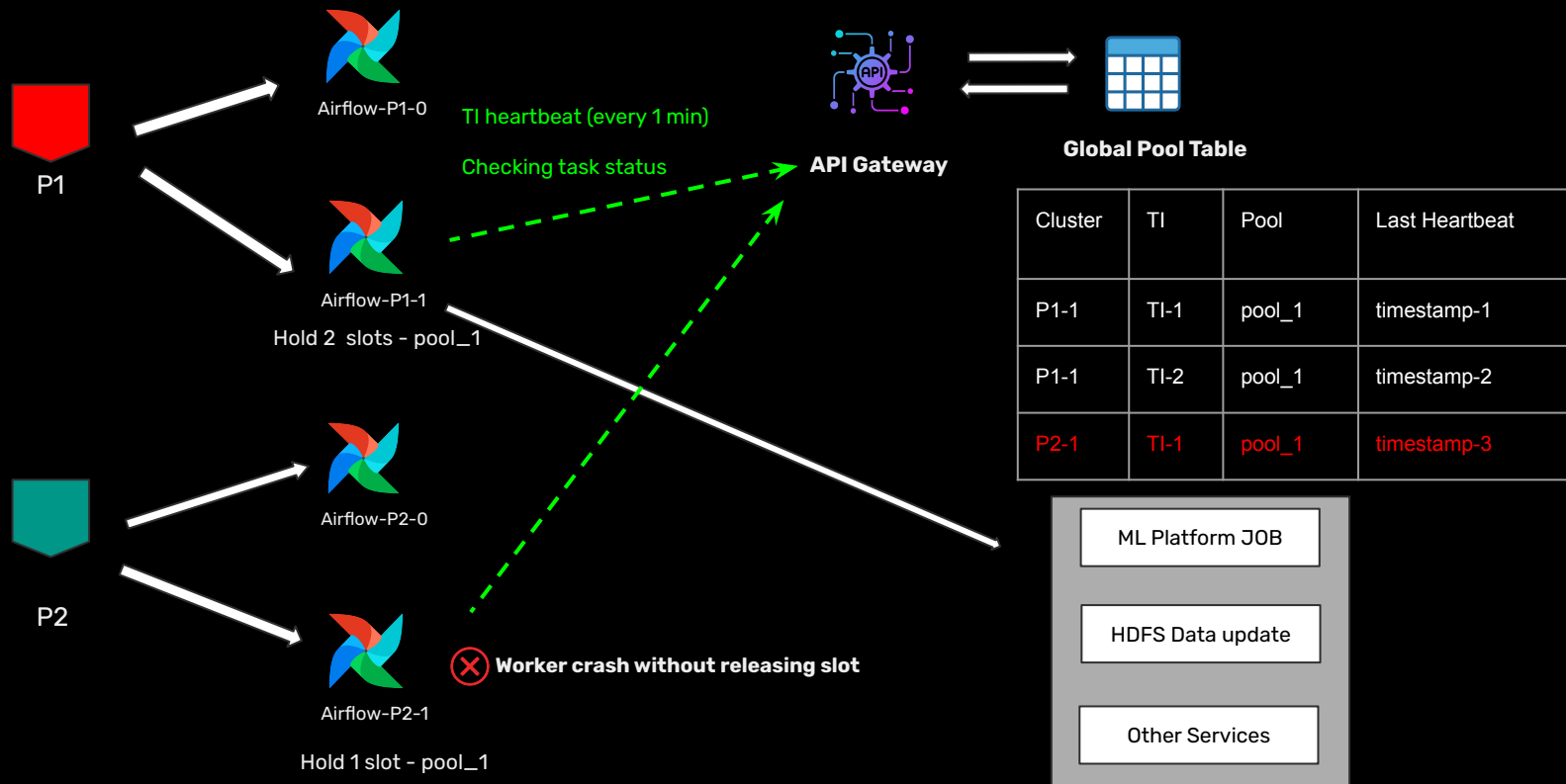
Global Pool - slot exception handling



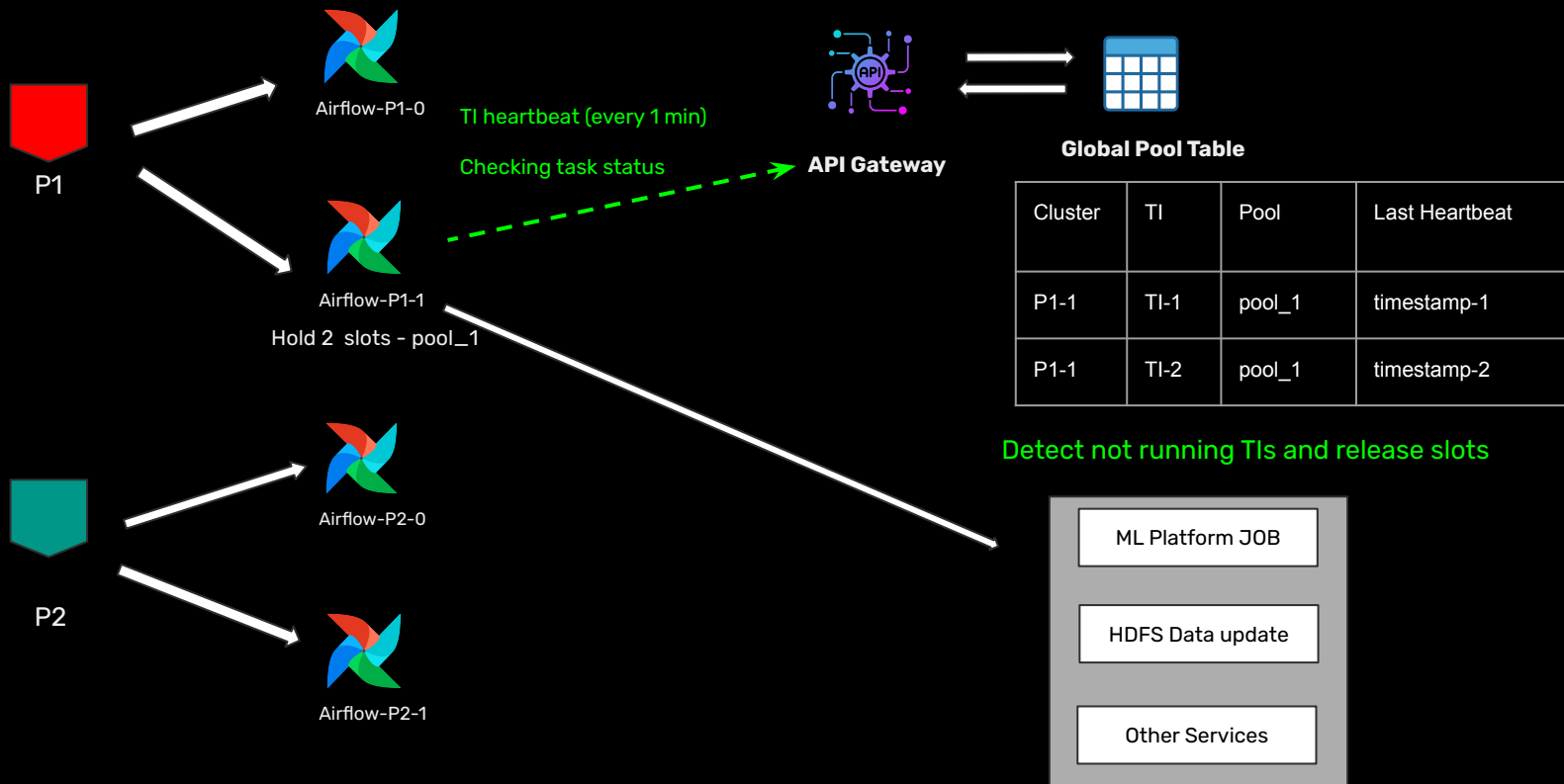
Global Pool - slot exception handling



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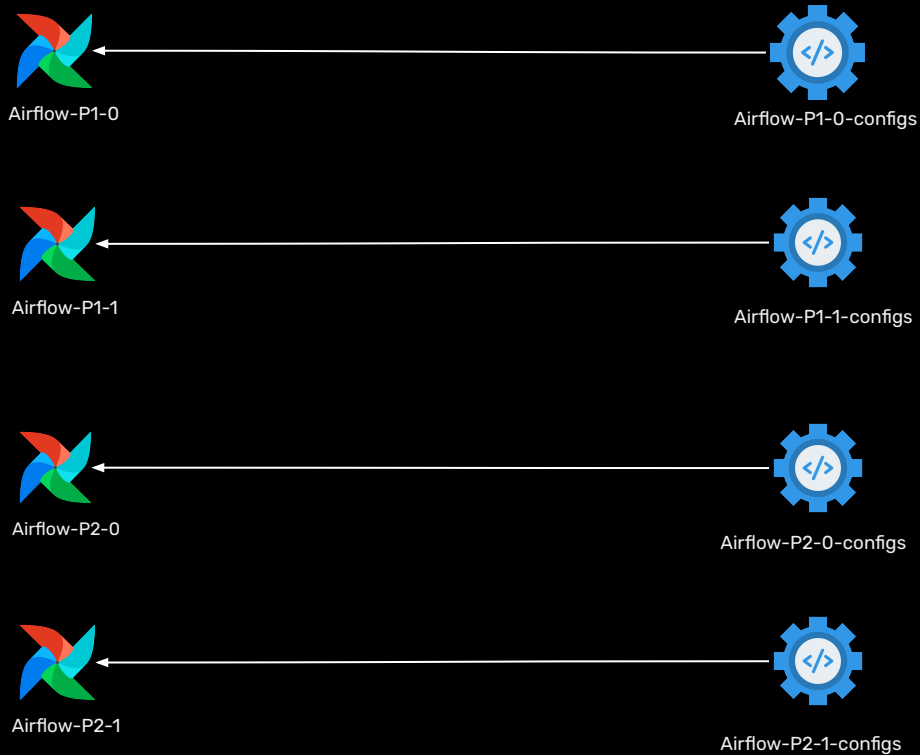


Global Pool - slot exception handling

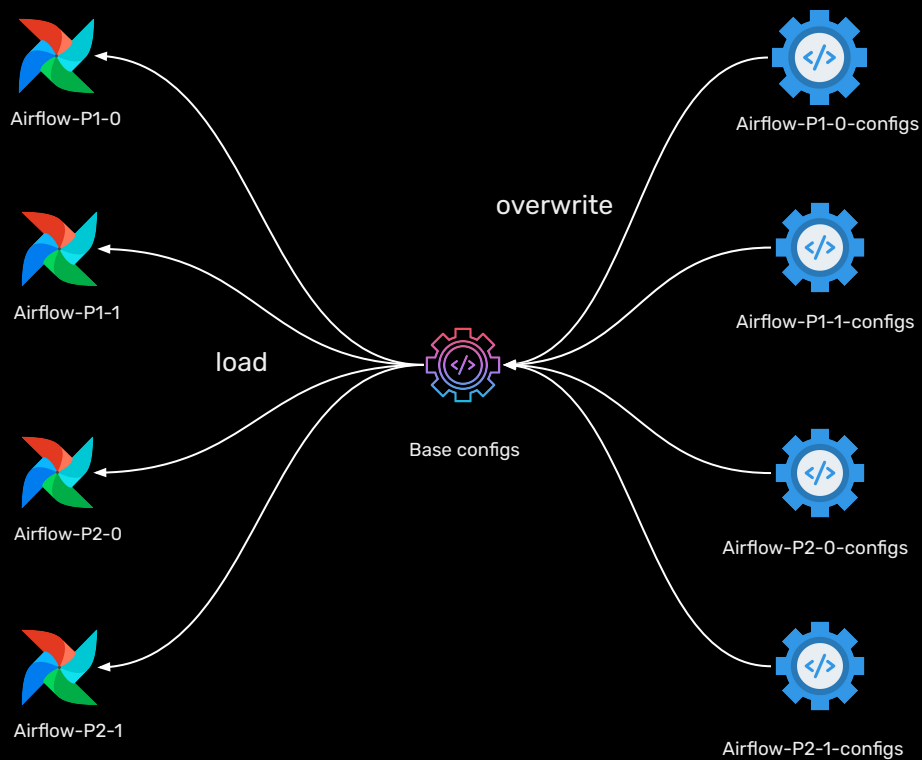


What's Next ?

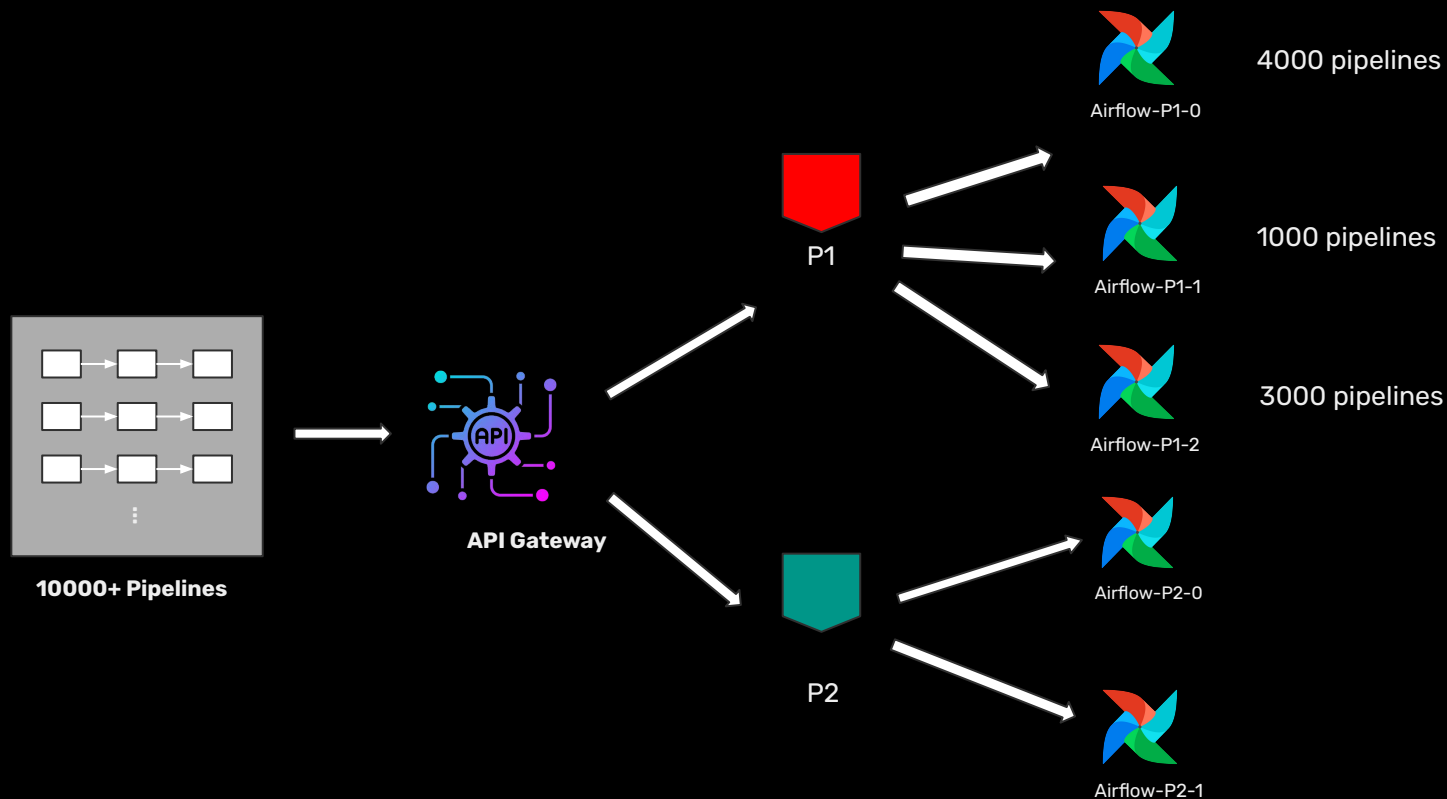
Next Step 1 - Federated config center



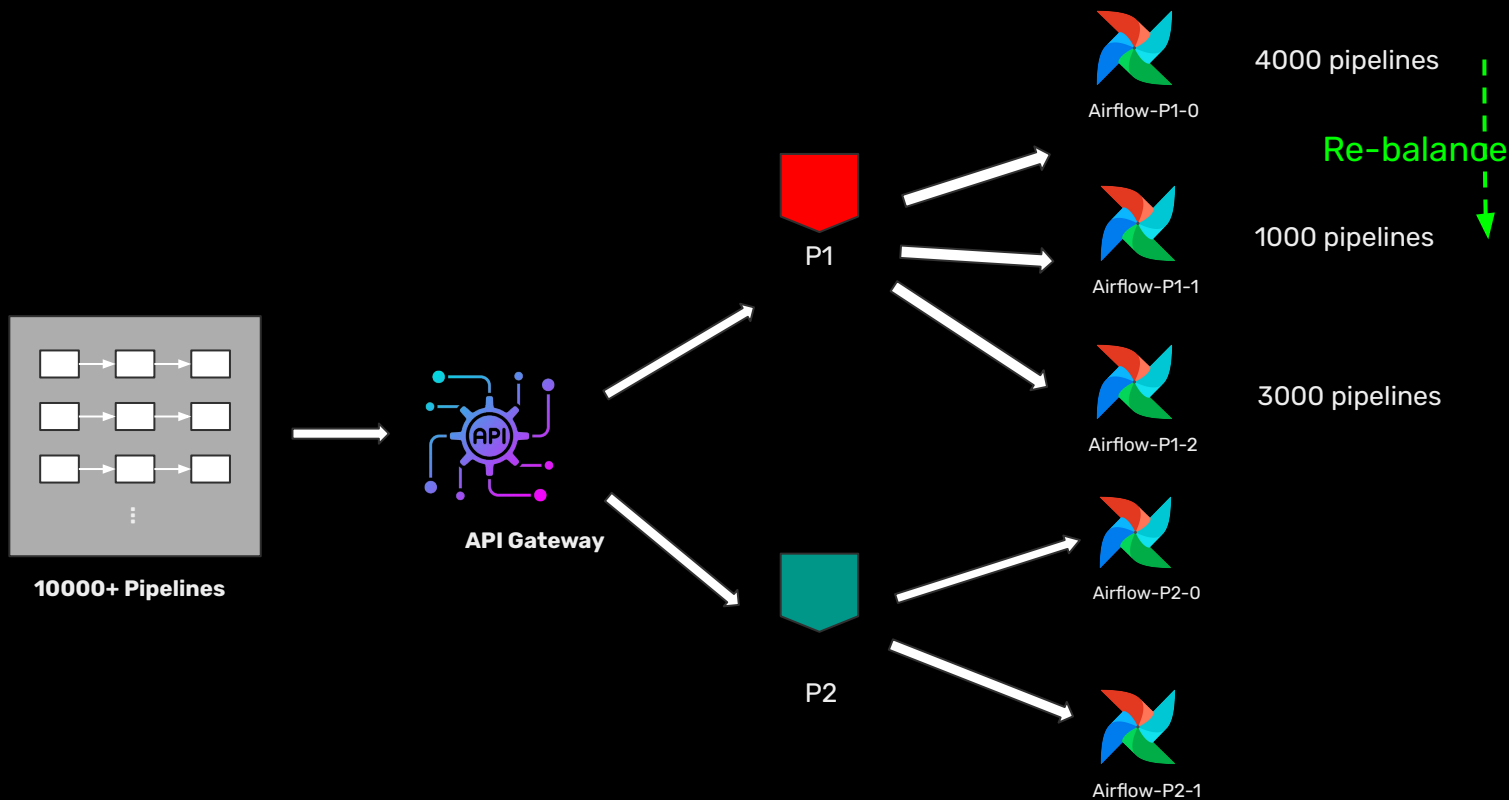
Next Step 1 - Federated config center



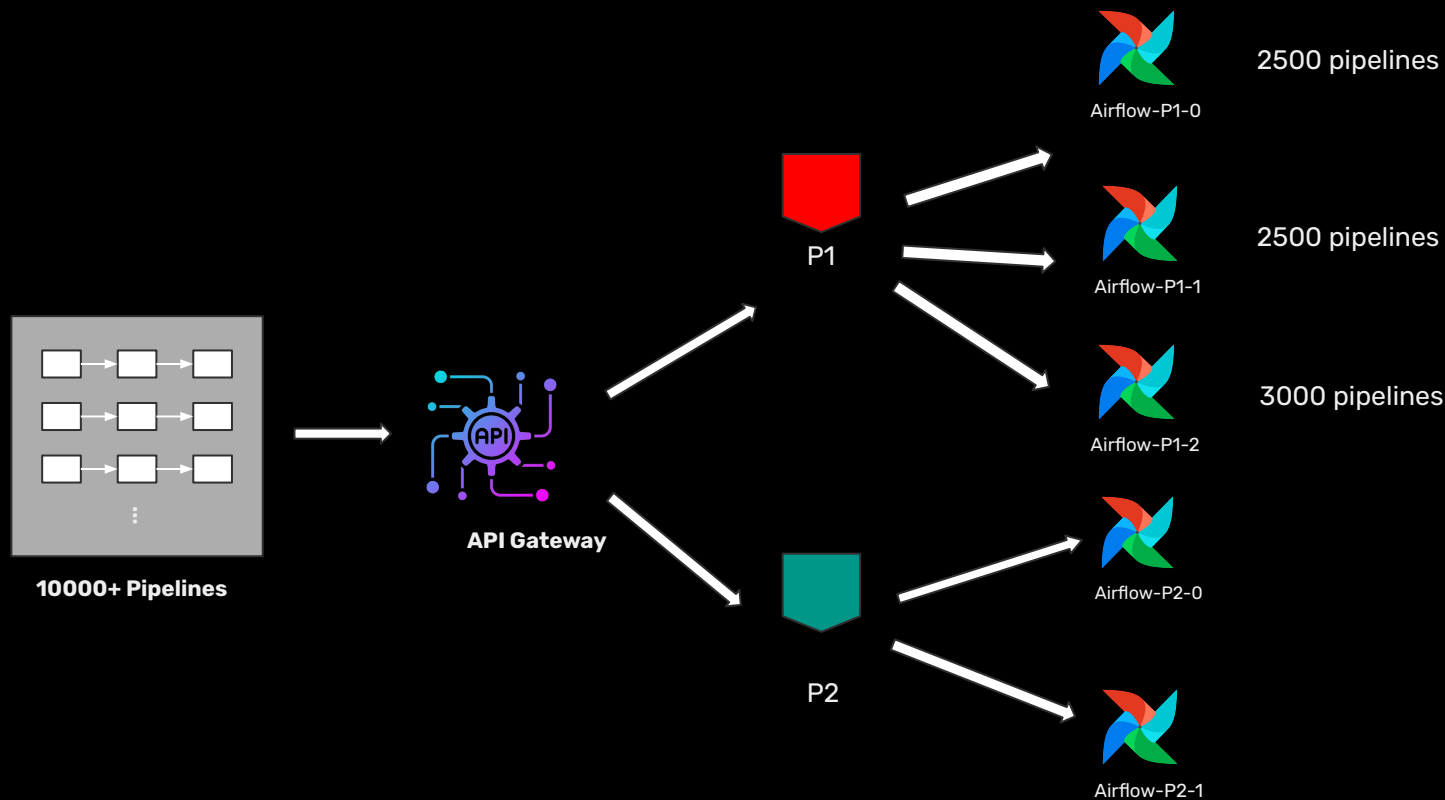
Next Step 2 - Dynamic Workload Re-balance



Next Step 2 - Automatic Workload Re-balance



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Q & A

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